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Domain-robust vision transformer with hierarchical swin encoding for explainable low-latency driver drowsiness detection

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Driver drowsiness is a significant cause of road accidents worldwide, often leading to fatalities due to delayed reaction times and loss of vehicle control. Traditional fatigue detection systems face several limitations, including poor generalization across various conditions, a lack of model transparency, and an inability to operate on low-power platforms. To address these issues, we propose Enhanced Multi-path Token-Fusion Vision Transformer (ViT), Light-VTD, a novel lightweight architecture designed for accurate and explainable fatigue detection from facial images. Unlike existing transformer-based methods, Light-VTD features a fused-MBConv block for efficient local feature extraction, a position-aware token mixer for capturing global context, and a multi-path token fusion mechanism that enhances spatial consistency across different resolutions. Furthermore, we adapt Gradient-weighted Class Activation Mapping (Grad-CAM) to generate interpretable attention heatmaps that align with clinically relevant fatigue indicators. Our experimental models are trained on a large-scale dataset comprising over 167,000 labeled images, compiled from four diverse public datasets: MRL Eye, nthuDDD2, and UTA-RLDD. The Light-VTD model achieves an intra-dataset accuracy of 99.2% and at least 93% in cross-dataset transfer, outperforming current ViT baselines by 2–4%. It supports inference with <120 ms latency on low-power devices (Raspberry Pi 4). Moreover, we have integrated the model within a web-based application that provides live predictions and visual explanations for fatigue monitoring. This study contributes a robust, interpretable, and deployable solution with potential applications in fleet-wide Advanced Driver Assistance Systems (ADAS), workplace safety platforms, and portable telehealth diagnostics.

Keywords Drowsiness detection, Vision transformer (ViT), Explainable AI (XAI), Token fusion, Embedded systems

Driver drowsiness is a major factor contributing to road traffic accidents worldwide. It leads to delayed reaction times, impaired situational awareness, and a loss of vehicle control. According to the World Health Organization, road traffic crashes result in approximately 1.3 million fatalities each year[1], with driver fatigue accounting for an estimated 20–25% of highway-related deaths[2,3]. In addition to safety concerns, fatigue-related incidents create noteworthy economic and societal burdens[4], particularly in sectors such as transportation, logistics, and emergency services, where prolonged working hours and limited monitoring are common. To address this issue,

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governments and regulatory bodies are increasingly mandating in-vehicle driver monitoring systems[5,6]. The European Union's General Safety Regulation (GSR) requires Driver Drowsiness and Attention Warning (DAW) systems in new vehicles, while similar initiatives are being adopted by the U.S. National Highway Traffic Safety Administration within ADAS[7,8].

Early approaches to detecting drowsiness relied on physiological signals, such as electroencephalography (EEG), electrooculography (EOG), and photoplethysmography (PPG). While these methods can capture subtle neurological and cardiovascular indicators of fatigue, they are often intrusive, sensitive to motion artifacts, and impractical for long-term use in real-world situations[9,10]. In contrast, vision-based approaches offer a non-invasive alternative by analyzing facial cues, such as eye closure, blinking patterns, yawning, and head pose. Traditional computer vision methods[11–13] that use handcrafted features-such as PERCLOS, the Eye Aspect Ratio (EAR), and the Mouth Aspect Ratio (MAR)-have shown promise but often perform poorly under varying lighting conditions, occlusion, head movement, and demographic diversity[14–16]. Deep neural networks have demonstrated superior performance on still image datasets, with architectures such as VGG, ResNet, and MobileNet often fine-tuned for binary alertness classification. However, many studies[17–19] are limited by small dataset sizes, leading to overfitting and poor generalization across varying real-world conditions. Another major concern is the lack of transparency in these models. Transfer learning models often function as black boxes, providing limited explanations for their predictions. This lack of interpretability poses major challenges in regulated fields such as transportation safety and clinical assessment. More recently, ViTs have appeared as a promising alternative. They can model long-range dependencies and global spatial relationships across image patches. While ViTs have achieved state-of-the-art results in general computer vision tasks, their application in drowsiness detection is still in its early stages.

Several critical gaps remain unaddressed in the current literature. Most existing approaches[20,21] are trained on the MRL Eye or nthuDDD2 datasets, without validating their performance on external corpora. As a result, these models tend to be highly sensitive to domain-specific biases and struggle to generalize to unseen samples. Moreover, due to their quadratic attention mechanisms, ViTs are not suitable for resource-constrained environments[22,23]. Many transformer-based models also lack optimization for low-latency performance, which restricts their use in wearable systems. Although ViTs inherently produce attention maps, these are seldom utilized effectively for post hoc explainability in drowsiness detection[24]. Current studies do not correlate model attention with physiologically meaningful regions (such as eyelid movement and periocular texture), which reduces trust in the model's decisions. This lack of interpretable output is a major barrier in safety-critical applications that require clinical or regulatory validation[25,26]. Additionally, most existing works rely on fundamental or arbitrary data augmentation strategies. Without an evolution from simple to complex visual transformations-like motion blur, occlusion, and low-light distortion-models are less likely to develop the visual resilience needed for robust performance.

There is also a general absence of systematic ablation studies assessing individual model components' contribution. This omission complicates the diagnosis of bottlenecks and hinders effective optimization of architectures. Finally, few studies benchmark models based on deployment metrics such as inference speed, power consumption, and frame-rate performance on embedded systems. Real-time fatigue monitoring requires accuracy and efficient inference pipelines that satisfy strict timing and energy constraints. The lack of such deployment-focused evaluations makes it challenging to determine the practical feasibility of current methods. Therefore, this study aims to develop a reliable, efficient, and user-friendly system for detecting driver drowsiness using camera-based image analysis. Our study sets out to achieve the following objectives:

- Design a model that can accurately identify signs of drowsiness from a diverse collection of facial images while remaining fast and lightweight enough for practical use in vehicles and portable devices.
- Use a step-by-step training strategy where the model is gradually exposed to increasingly complex examples, allowing it to learn more effectively and become more robust.
- Assess the system's ability to generalize accurately and provide visual explanations that indicate the parts of the face the system focuses on when making predictions for users and regulators.
- Build a working demonstration in the form of a web application that shows predictions and explanations quickly, even on low-powered devices like small computers used in cars.

To achieve these objectives, this study proposed Light-VTD for real-time drowsiness classification. Figure 1 outlines the Light-VTD workflow, highlighting its efficiency, robustness, and interpretable decision-making. We describe the multi-dataset approach and the preprocessing pipeline, which includes resizing, ROI localization/cropping, contrast-limited adaptive histogram equalization (CLAHE), normalization, and optional temporal sub-sampling. This reduces nuisance variability before model inference. The evaluation protocol employs metrics like F1 score, Matthew's Correlation Coefficient (MCC), and Precision-Recall Area Under the Curve (PR-AUC), along with repeated runs and statistical checks for baseline comparisons and performance attribution. Lastly, we connect model outputs to trust and deployment with Grad-CAM localization over periocular regions and a lightweight web interface. This shows that Light-VTD enables real-time inference with spatially grounded explanations, suitable for safety-critical monitoring. The proposed Light-VTD model features a lightweight fused-MBConv stem for local feature extraction, a position-aware token mixer for global context modeling, and a multi-path token fusion module for capturing hierarchical spatial relationships. Training involves class-weighted loss with label smoothing and stochastic depth regularization. Key contributions of this study are as follows:

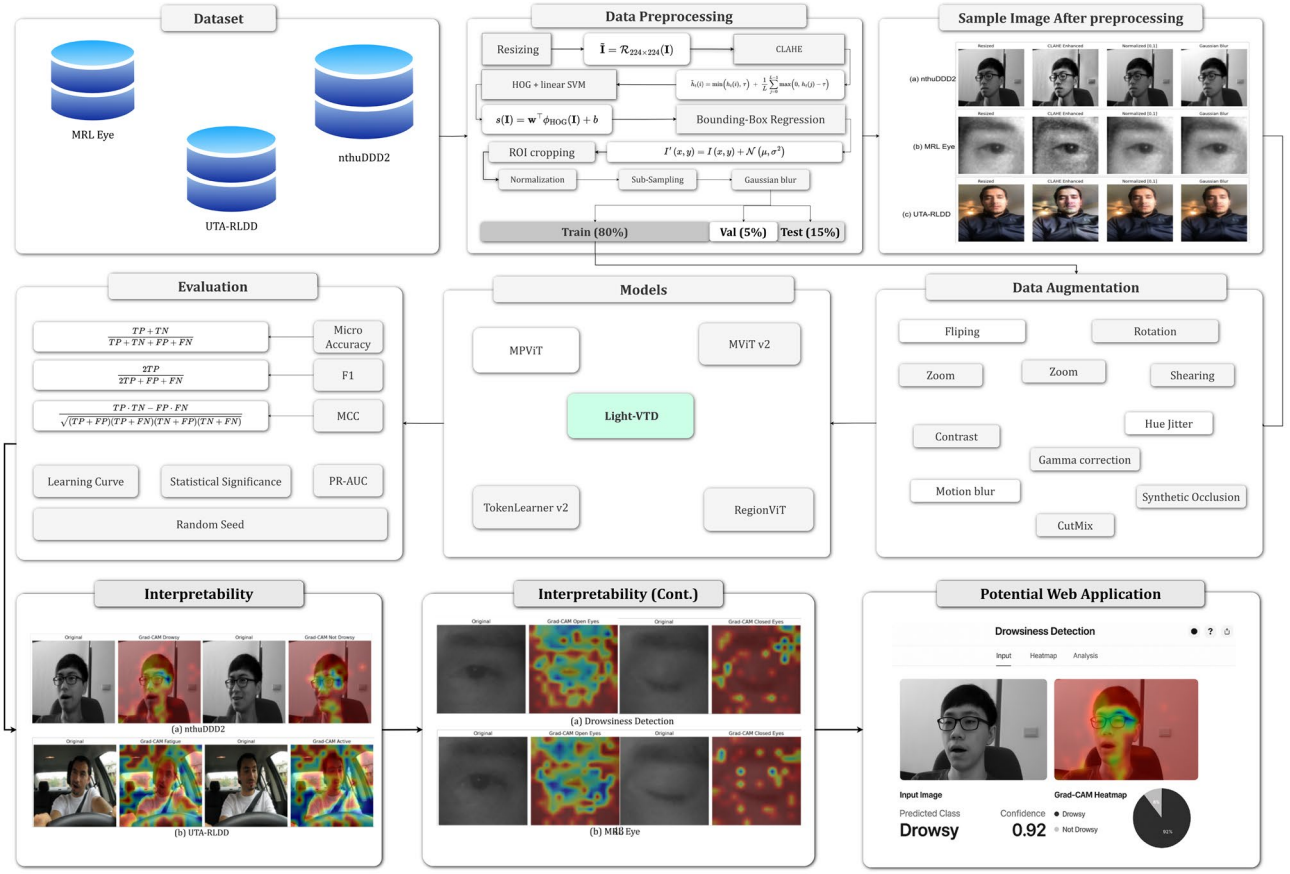


Fig. 1. End-to-end workflow and experimental design of the proposed Light-VTD framework. It summarizes the datasets employed, the preprocessing and ROI standardization pipeline, and representative examples illustrating the effect of contrast normalization and region-focused enhancement. We further depict the evaluation protocol and metric suite used for rigorous performance assessment, the comparative modeling setup highlighting Light-VTD against transformer baselines, and the structured data augmentation curriculum spanning geometric, photometric, and temporal perturbations. The final components illustrate the interpretability and deployment aspects of the framework, where Grad-CAM visualizations confirm that predictions are driven by periocular/facial cues, and a lightweight web interface demonstrates efficient inference with transparent explanations..

- We introduced a novel multi-path token fusion module within Light-VTD that adaptively aggregates spatial information across different window scales, enhancing feature consistency and improving classification of subtle fatigue cues such as micro-closures.
- We implemented structured visual complexity through incremental data augmentation stages, enabling the Light-VTD to develop robust, domain-invariant features. This improves generalization across different datasets and reduces confusion between similar classes.
- Validation using leave-one-dataset-out testing showed that the model achieves an accuracy of 93% or higher on unseen datasets, demonstrating strong generalization capabilities. Wilcoxon signed-rank tests confirmed it outperforms five leading ViT baselines by 2–4%.
- The model used transformer attention blocks to generate class-specific saliency maps that align with visible fatigue cues. We launched a production-ready Flask-based web interface that provides predictions and heatmaps in under 120 ms, while achieving frame rate of 7.9 FPS.

This paper is structured as follows: “Related works” reviews existing techniques and recent improvements in Transformer-based models. “Methodology” outlines the proposed methodology, covering dataset preparation, curriculum-based augmentation, the Light-VTD architecture, and training protocols. “Result analysis” details the experimental setup and evaluates the model’s performance across intra- and cross-dataset settings using various metrics, ablation studies, and robustness tests. “Discussion” discusses key findings, including performance improvements, generalization, practical implications, and limitations. Finally, “Conclusion” summarizes the contributions and suggests future real-world fatigue monitoring systems directions.

Related works

Physiological signal-based approaches

Early studies on drowsiness detection primarily relied on physiological signals, which capture neurological and ocular indicators of fatigue. Awais et al.[27] proposed a hybrid EEG–ECG framework that reduced the number of electrodes while achieving 80.9% accuracy in a simulated driving environment. However, its reliance on controlled settings and intrusive sensors limits real-world applicability. Liu et al.[28] employed EEG and EOG features with a fuzzy SVM classifier, reporting 95.11% accuracy on the SEED-VIG dataset and integrating an IoT-based warning mechanism. Despite strong performance, signal instability and motion artifacts pose challenges in mobile driving scenarios. Kołodziej et al.[29] analyzed blink-related EOG signals during office tasks, achieving up to 93% accuracy in user-dependent settings, though limited subject diversity restricted generalizability. Sharma et al.[30] introduced a wavelet–Tsallis entropy model using EEG, EMG, and EOG data from a large cohort, but the approach suffered from high computational cost and reduced sensitivity to imbalanced fatigue stages. While physiological approaches offer high sensitivity under controlled conditions, their intrusive nature, susceptibility to noise, and limited scalability hinder deployment in real-world driver monitoring systems.

Vision-based approaches

Vision-based methods provide a non-intrusive alternative by analyzing facial cues such as eye closure, blinking frequency, yawning, and head pose. Spandana et al.[31] utilized MobileNet-SSD for eye-state detection, achieving 97% accuracy on a small-scale dataset, but lacked validation under diverse environmental conditions. Albadawi et al.[32] combined handcrafted facial features with a Random Forest classifier, reporting 99% accuracy on nthuDDD2; however, performance degraded under low illumination and occlusion due to reliance on HOG-based face detection. Safarov et al.[33] focused on eye gaze and head movement analysis in video sequences, achieving strong head-lean detection but struggling with sunglasses and occlusions. Li et al.[34] employed multi-feature fusion with transfer learning to reach real-time performance, though robustness declined under poor lighting and facial accessories. Classical feature-based approaches such as that of Owen et al.[35] demonstrated moderate accuracy in controlled environments but failed to generalize across subjects. Akrout et al.[36] extended facial cue analysis to multiple datasets, achieving around 96% accuracy, yet reported false alarms caused by head motion and illumination variability. These studies highlight the practicality of vision-based systems but also reveal limitations in robustness, generalization, and interpretability when deployed in unconstrained environments.

Deep convolutional neural networks (CNNs) have significantly improved drowsiness detection by learning discriminative facial representations directly from data. Almazroi et al.[37] proposed a MobileNet-based multi-task system for drowsiness and distraction detection, achieving high accuracy and F1 scores, though evaluation was restricted to favorable lighting conditions. Vijaypriya et al.[38] introduced a multi-channel CNN optimized via Flamingo Search Optimization, reporting strong performance on benchmark datasets but lacking real-world validation. Li et al.[39] designed a lightweight CNN for wearable glasses, achieving 93.9% blink detection accuracy, albeit on a small dataset with limited subject diversity. Venkateswarlu et al.[40] proposed a similar CNN-based multi-alert framework, but robustness under adverse weather and nighttime conditions remained untested. Florez et al.[41] evaluated multiple CNN backbones for eye-state classification, achieving near-perfect accuracy on the NITYMED dataset, though the limited number of video samples constrained generalizability. Despite strong local feature extraction, CNN-based methods often struggle with capturing long-range spatial dependencies and maintaining robustness across datasets with varying capture conditions.

ViTs have recently emerged as a powerful alternative by modeling global spatial relationships through self-attention mechanisms. Jarndal et al.[42] applied ViTs to facial images and achieved high accuracy on nthuDDD2 and UTA-RLDD, but performance degraded under severe occlusion. Xiao et al.[43] combined handcrafted facial parameters with a residual Swin Transformer, achieving competitive accuracy at the cost of a large model size that limits edge deployment. Hassan et al.[44] integrated ViT and Swin Transformer architectures for real-time detection, demonstrating strong accuracy across multiple datasets but incurring high computational overhead. Gao et al.[45] proposed a CLIP-based transformer framework, introducing semantic alignment at the expense of handcrafted prompts and reduced efficiency. El-Nabi et al.[46] combined diffusion models with Swin Transformers to enhance robustness, achieving state-of-the-art accuracy but with prohibitive computational cost for low-power devices. Zhang et al.[47] explored transformer-based fatigue detection using EEG signals, presenting methodological novelty but limited real-time applicability. While ViT-based models improve global context modeling, most existing approaches remain computationally expensive for deployment in resource-constrained environments.

Explainable and real-time systems

With the increasing deployment of driver monitoring systems in safety-critical environments, explainability and real-time efficiency have become essential requirements alongside accuracy. Explainable AI (XAI) aims to provide transparent decision evidence by identifying which visual or physiological cues drive a model's prediction, improving user trust, regulatory acceptance, and clinical interpretability[48]. In the specific context of drowsiness and fatigue detection[49], explainability is particularly important for verifying whether models attend to physiologically meaningful regions such as the eyelids, periocular area, and facial muscle tension, rather than spurious correlations[50].

Recent studies primarily employ post hoc[51] explainability layered on top of predictive models. Iglesias et al.[50] investigated driver drowsiness detection with XAI and emphasized interpretable outputs to support human-understandable decisions; however, many such XAI integrations remain qualitative and are not systematically validated for cross-domain consistency. Similarly, Kiruba and Sofia[49] presented an XAI-based driver fatigue detection framework to promote safer driving, yet common limitations in this line of work include limited

external validation and incomplete reporting of how explanations correlate with clinically relevant fatigue cues. More broadly, Alharthi et al.[48] presented XAI for sensor-signal interpretation in health monitoring and highlighted recurring challenges that also apply to explanation faithfulness, robustness under noise, and the need for robust evaluation protocols. Consistent with these observations, several works apply saliency mapping or attention visualization as interpretability proxies without quantitative validation or cross-dataset consistency analysis[52]. Further, Chengula et al.[53] used XAI to enhance driver anomaly detection and argued that interpretable decisions can support actionable alerts; nevertheless, many attention-guided studies[54] still do not explicitly demonstrate alignment between highlighted regions and physiologically grounded fatigue indicators. Wearable and sensor-driven XAI systems for driving-related cognitive monitoring have also been explored[55], but these approaches often shift the challenge toward sensor reliability and user compliance. Moreover, yagin et al.[56] targetted fatigue-related conditions outside driving domains, reinforcing the general observation that interpretability is increasingly expected, yet domain-specific physiological validation remains underdeveloped.

Real-time deployability remains a major barrier for fatigue monitoring on embedded platforms. Facial analysis pipelines can achieve operational frame rates by using shallow backbones, reduced input resolution, or simplified feature extraction[57–59]. However, these optimizations may reduce robustness under occlusion, illumination changes, or cross-domain shifts, which are common in real driving environments. Transformer-based models[60,61] often incur higher computational cost due to self-attention operations, creating deployment challenges without specialized acceleration. Efficiency-oriented designs attempt to mitigate this issue, such as hybrid ViT models[62–64] with attention mechanisms designed to reduce redundancy while improving interpretability, and optimized deep hybrid learning approaches for multi-channel EEG-based drowsiness detection[65]. Nonetheless, deployment evaluations are frequently limited to simulated conditions and often omit critical reproducibility details such as power consumption, operating modes, and hardware configurations. Furthermore, while hardware-oriented studies demonstrate feasibility in non-driving fatigue domains[66], translating these insights into robust in-vehicle monitoring requires explicit validation under realistic conditions and transparent reporting of trade-offs.

Existing systems tend to treat explainability and real-time efficiency as separate design objectives. Comprehensive reviews[67–69] of fatigue detection methodologies present that the field still lacks unified frameworks that jointly optimize generalization, interpretability, and deployment efficiency with transparent evaluation protocols. Recent XAI frameworks in adjacent healthcare analytics similarly stress the importance of interpretable yet robust modeling under practical constraints[70,71]. These gaps motivate the development of lightweight architectures that embed explainability into the model design and evaluation pipeline while maintaining low-latency inference suitable for edge deployment. In this context, lightweight hybrid ViT models with integrated feature explanations represent a promising direction for practical and trustworthy driver drowsiness monitoring systems.

Methodology

Data description

We used three publicly available datasets to capture fatigue-related visual patterns in both static and dynamic conditions, enabling us to create adaptable deep learning models for real-world scenarios. The first dataset, MRL Eye Dataset[72] includes 84,898 high-resolution images-41,946 with closed eyes and 42,952 with open eyes-captured in various lighting conditions and from different camera angles and subjects. This diversity improves the model's robustness, allowing it to adapt to changing environments. It is crucial in aviation and logistics, where operators experience extended periods of strain and variable lighting. Sample images from the experimental datasets are shown in Fig. 2.

To incorporate dynamic visual cues, for second dataset, we utilized the nthuDDD2[44], which contains 66,521 labeled video frames from real driving sessions. Of these, 36,030 frames are marked as "Drowsy" and 30,491 as "Not Drowsy." This video-based dataset helps models learn key fatigue indicators such as yawning and slow eye blinks, essential for real-time driver assistance systems that monitor alertness and prevent accidents. Lastly, the UTA-RLDD (Real-Life Drowsiness Dataset)[73] includes 11,787 facial sequences labeled as either Fatigue (4557) or Active (7230). Captured in uncontrolled environments, it features challenges like occlusion and variable lighting. This dataset's realistic conditions are perfect for training models for practical applications like in-car systems and mobile fatigue detection, ensuring effective real-time context-aware performance.

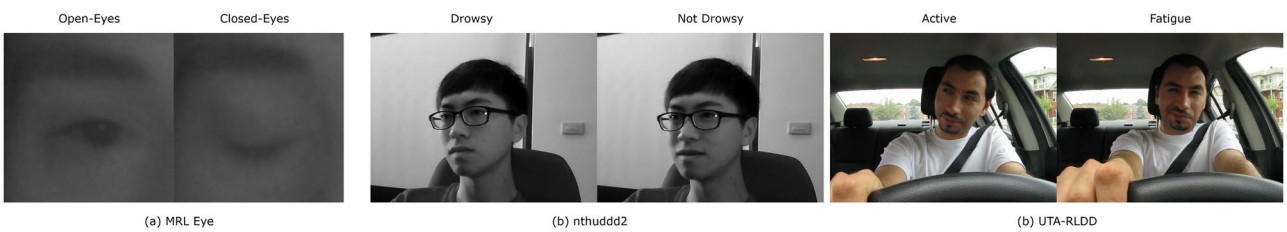


Fig. 2. Sample image from each class of both (**a**) MRL eye, (**b**) nthuDDD2, and (**c**) RLDD datasets.

Data preprocessing

All face images were standardized to a resolution of 224×224 prior to tokenization. To improve visibility of subtle ocular boundaries under uneven illumination, we applied CLAHE. It improves local contrast around the periocular region, which is critical for distinguishing partial eyelid closure under challenging lighting. We verify gains via ablation (Table 11). For region localization, we used a cascaded detection pipeline consisting of HOG+linear SVM followed by RetinaFace for non-frontal faces and occlusion handling[74]. The cascaded HOG–RetinaFace pipeline is applied only during offline preprocessing; runtime inference operates solely on cropped facial regions. Given an anchor box $a = \{x_a, y_a, w_a, h_a\}$ and predicted offsets $\Delta = \{\delta_x, \delta_y, \delta_w, \delta_h\}$, the refined face box is computed as:

$$b = \{x_a + \delta_x w_a,\ y_a + \delta_y h_a,\ w_a e^{\delta_w},\ h_a e^{\delta_h}\}.$$

The extracted facial ROI preserves upper-face cues (e.g., eyelid aperture and gaze direction) that are highly discriminative for fatigue inference. Finally, all cropped ROIs were normalized to the $[0, 1]$ range. For video-based datasets (nthuDDD2 and UTA-RLDD), we reduced redundancy via temporal sub-sampling (every k-th frame) and generated overlapping windows to preserve short-term behavioral dynamics. Figure 3 shows an example of the final preprocessed output used by the transformer backbone. Here, CLAHE differs from global histogram equalization by operating on small local tiles and incorporating contrast limiting. This approach prevents the amplification of noise while enhancing the visibility of features such as partial eyelid closure, iris boundaries, and periocular texture. This enhancement is particularly useful for low-light or low-contrast images[75], like those in the MRL Eye dataset, where signs of fatigue may be difficult to distinguish. Results indicate that our preprocessing pipeline consistently enhances the PR-AUC on the other two datasets, highlighting its effectiveness in retaining critical ocular features for accurate drowsiness detection.

Data augmentation

Preprocessing ensures input consistency, while data augmentation enhances the training distribution. Therefore, they are handled separately. A structured augmentation strategy was applied solely to the training data to boost robustness and minimize overfitting, leaving the validation and test sets unchanged. Augmentation occurred in three stages: (1) geometric transformations, (2) photometric changes, and (3) complex distortions. Motion blur was applied after spatial and photometric transformations to maintain facial structure before adding sensor and motion artifacts. CutMix was utilized last in the curriculum after applying motion blur and Cutout, combining each augmented image with a second randomly selected training image. The mixing coefficient λ was derived from a Beta distribution, and the training label was adjusted using the formula $\tilde{y} = \lambda y_i + (1 - \lambda) y_j$. This sequence ensures that CutMix is applied to already modified samples, acting as a distribution-level regularizer and preventing early label noise, while ensuring geometric and photometric realism before mixing regions from different samples. For temporal datasets, frame sub-sampling and temporal jittering were included to enhance

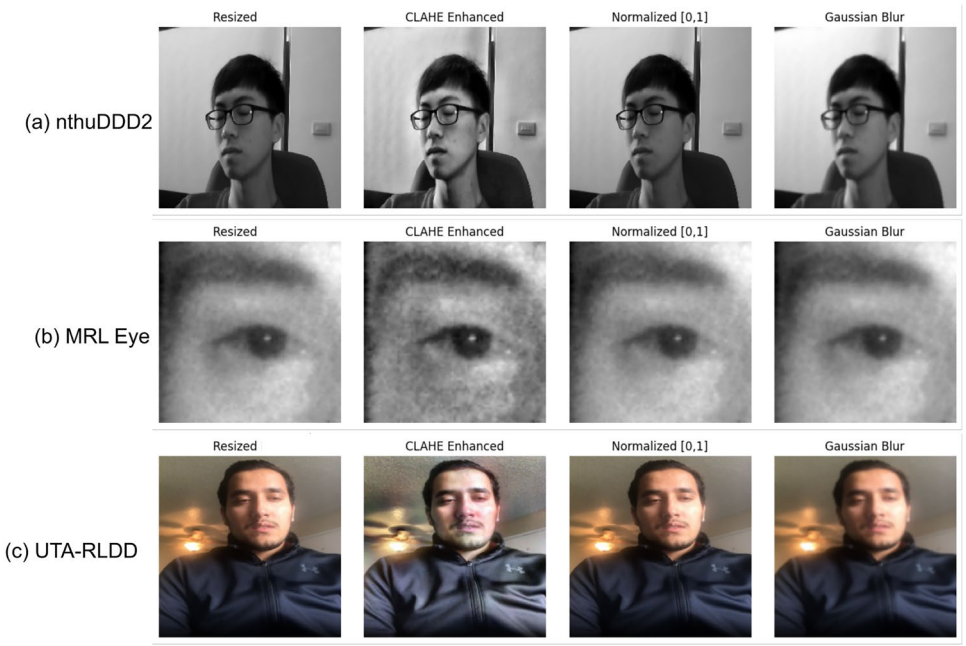


Fig. 3. Visual illustration of preprocessing steps across datasets. Examples from (**a**) nthuDDD2, (**b**) MRL eye, and (**c**) UTA-RLDD are shown after resizing, CLAHE enhancement, normalization, and Gaussian blur. CLAHE enhances local contrast around the eyelid and periocular regions, improving the visibility of fatigue-related cues under uneven illumination, while subsequent normalization and blur standardize intensity ranges and simulate sensor-induced degradation for robustness training.

Dataset	Augmentation type	Value range	Final value
MRL eye	Vertical flip	Probability: 0–1	0.3
	Rotation	$\pm 0^\circ$–25°	$\pm 15^\circ$
	Contrast shift	±0%–40%	±25%
	Hue jitter	±0%–20%	±12%
	Zoom scaling	0.85×–1.15×	1.05×
nthuDDD2	Horizontal flip	Probability: 0–1	0.6
	Rotation	$\pm 0^\circ$–35°	$\pm 22^\circ$
	Temporal frame skip	1–6 frames	4 frames
	Motion blur	Kernel: 1–7 px	5 px
	Synthetic occlusion	0%–30% of face area	15%
UTA-RLDD	Horizontal flip	Probability: 0–1	0.4
	Rotation	$\pm 0^\circ$–28°	$\pm 16^\circ$
	Gamma correction	$\gamma \in [0.7, 1.5]$	1.2
	Shear distortion	Factor: 0–0.5	0.25
	Random patch drop	0%–15% of region	8%

Table 1. Summary of augmentation strategies, parameter ranges, and selected values for training across datasets.

Dataset	Class	Original count	Train (80%)	Target ×2	Target ×4	Target ×6	Val (5%)	Test (15%)
MRL Eye	Closed Eyes	41,946	33,556	67,112	134,224	201,336	2097	6293
	Open Eyes	42,952	34,362	68,724	137,448	206,172	2148	6442
	Total	84,898	67,918	135,836	271,672	407,508	4245	12,735
nthuDDD2	Drowsy	36,030	28,824	57,648	115,296	172,944	1802	5404
	Not Drowsy	30,491	24,393	48,786	97,572	146,358	1525	4573
	Total	66,521	53,217	106,434	212,868	319,302	3327	9977
UTA-RLDD	Fatigue	4557	3645	7290	14,580	21,870	228	684
	Active	7230	5784	11,568	23,136	34,704	362	1084
	Total	11,787	9429	18,858	37,716	56,574	590	1768

Table 2. Class distribution and data splits across drowsiness detection datasets.

robustness against speed variations and motion-induced noise. Table 1 outlines the types of augmentations, parameter ranges, and final values used for each dataset.

Each dataset was divided using stratified sampling into training (80%), validation (5%), and testing (15%) subsets to preserve class distributions. To mitigate imbalance and increase intra-class variability, the training set was expanded at three levels (×2, ×4, and ×6), while validation and test sets remained unchanged. Table 2 reports the class-wise allocation and the resulting augmented training sizes.

Baseline models

We compare the proposed Light-VTD with four representative ViT backbones that represent complementary design choices: multi-path aggregation (MPViT), hierarchical multiscale token pyramids (MViT v2), adaptive token reduction (TokenLearner ViT), and region-aware token interaction (RegionViT). All baseline models are trained and evaluated using the same data splits, input resolutions, and training protocols as Light-VTD. MPViT is an efficient multi-path ViT that aggregates information across multiple receptive-field scales through parallel feature streams. This approach is particularly relevant for capturing both fine periocular cues and broader facial context[76]. We use MPViT as the frame-level backbone for drowsiness classification and report its performance under the same training protocol as Light-VTD. MViT v2 constructs a hierarchical token pyramid with multiscale attention, striking a balance between accuracy and computational efficiency by progressively reducing spatial resolution while increasing channel capacity[77]. This multiscale hierarchy serves as a strong baseline for subtle facial dynamics, as it preserves both local detail and global structure. TokenLearner ViT adaptively selects a compact set of informative tokens, reducing redundancy and minimizing background noise[78,79]. This mechanism is relevant for driver monitoring because it prioritizes salient facial regions while lowering attention costs. RegionViT introduces complementary local and regional token streams with region-to-local interaction, aiming to retain fine details while incorporating stable coarse context[80,81]. We include RegionViT to benchmark region-aware modeling against the region-focused design choices in Light-VTD.

Proposed Light-VTD

Design objectives and scope

Light-VTD is a deployment-oriented drowsiness detector that is designed to (i) preserve fine periocular cues even under challenging lighting conditions, (ii) efficiently capture multi-scale facial context, and (iii) produce explanation maps suitable for safety-critical monitoring. Specifically, Light-VTD combines a lightweight convolutional stem for capturing local facial structures with a hierarchical transformer encoder for contextual reasoning. It also integrates an adaptive multi-path fusion module to combine complementary spatial scales without adding significant computational overhead. The architecture is applied to cropped face ROIs at a fixed input size of 224×224 pixels and outputs a dataset-specific binary label, while maintaining a consistent backbone across different benchmarks. As shown in Fig. 4 and Algorithm 1), the model utilizes a five-stage pipeline for facial analysis. The process begins with a MobileOne-derived convolutional stem that extracts local facial features such as eyelid boundaries, iris edges, and periocular textures. Next, it employs patch embedding to convert these features into tokens. Following this, hierarchical Swin encoding facilitates efficient contextual interactions among tokens and progressive patch merging. The fourth stage involves adaptive multi-path token fusion, which aggregates multi-scale token representations using learnable weights to enhance the model's understanding. Finally, a lightweight classification head, which consists of LayerNorm, global pooling, and a fully connected layer, generates the final prediction.

Notation

Let the preprocessed facial ROI be $I \in \mathbb{R}^{H\times W\times 3}$ with $H = W = 224$. The model produces logits $\mathbf{z} \in \mathbb{R}^C$ and class probabilities $\hat{\mathbf{y}} = \mathrm{Softmax}(\mathbf{z})$, where $C = 2$ for all datasets used in this study.

Local structure with low overhead

To capture subtle fatigue cues that are easily degraded by illumination and compression artifacts, Light-VTD begins with a compact convolutional stem inspired by MobileOne-style efficient designs. The stem uses a fused MBConv-like pattern to encode local edges and textures prior to tokenization. Given input I, the stem feature map is computed as:

$$F_{\mathrm{mob}} = \mathrm{BN}(\mathrm{Conv}_{3\times 3}(\phi(\mathrm{Conv}_{1\times 1}(I)))), \tag{1}$$

where $\phi(\cdot) = \mathrm{ReLU}(\cdot)$ and $F_{\mathrm{mob}} \in \mathbb{R}^{d_1\times H_1\times W_1}$. This stage provides locality and noise tolerance while avoiding the high cost of applying self-attention at the raw pixel level.

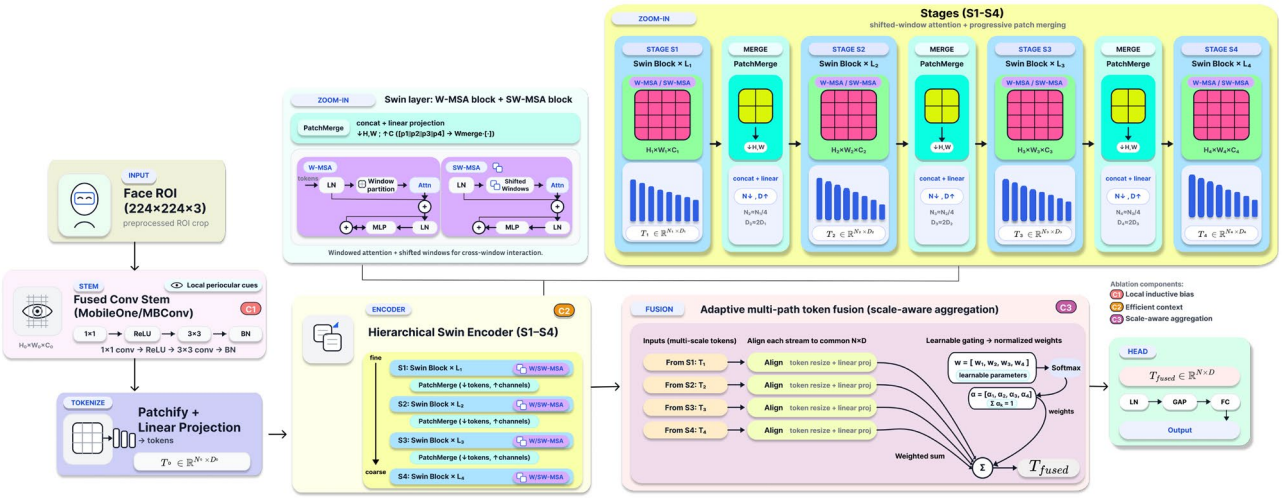


Fig. 4. Overview of the proposed Light-VTD architecture with ablation-mapped contributions and zoom-in details. Starting from a preprocessed face ROI, Light-VTD extracts local periocular features using a lightweight fused convolutional stem. The resulting feature map is divided into patches and linearly projected to form initial tokens $T_0 \in \mathbb{R}^{N_0\times D_0}$. These tokens are then processed by a hierarchical Swin transformer backbone distributed across four stages (S1–S4). In each stage, Swin blocks with Window-based Multi-Head Self-Attention (W-MSA) and Shifted Window-based Multi-Head Self-Attention (SW-MSA) are applied. This efficient contextual modeling is complemented by PatchMerge operations, which progressively reduce the token resolution while increasing the channel capacity (for example, $N_{i+1} \approx N_i/4$ and $D_{i+1} \approx 2D_i$). The multi-scale token outputs $\{T_1, T_2, T_3, T_4\}$ are fed into the proposed adaptive multi-path token fusion module. Each stream is aligned to a common token shape through token resizing and linear projection. Learnable weights $w = [w_1, w_2, w_3, w_4]$ are normalized using a Softmax function to obtain α_k, after which the fused representation is calculated. Finally, a classification head predicts the drowsiness state. Zoom-in insets illustrate the internal structure of a Swin layer and the PatchMerge mechanism, while the legend links C1–C3 to the corresponding ablation components.

Tokenization of local facial features
The stem output is partitioned into non-overlapping patches and linearly projected into an initial token sequence:

$$T_0 = \mathrm{Linear}(\mathrm{Partition}(F_{\mathrm{mob}})) \in \mathbb{R}^{N \times C_0}, \tag{2}$$

where N is the number of patches and C_0 is the embedding dimension. Compared to direct image patchification, tokenizing the stem features retains fine periocular structure while reducing sensitivity to background pixels and sensor noise.

Efficient contextual reasoning
Light-VTD uses four hierarchical transformer stages to model increasingly global facial context. Each stage reduces spatial resolution via patch merging while increasing channel capacity, enabling multi-scale reasoning without quadratic cost in token length. For a stage transition, four neighboring tokens are concatenated and projected:

$$P_{\mathrm{merge}} = W_{\mathrm{merge}} \cdot [p_1 \| p_2 \| p_3 \| p_4], \tag{3}$$

where $W_{\mathrm{merge}} \in \mathbb{R}^{4C \times 2C}$ is learnable and $\|$ denotes concatenation. Within each stage, tokens are processed by SW-MSA followed by an MLP with residual connection:

$$T_s \leftarrow \mathrm{MLP}(\mathrm{SW-MSA}(T_s)) + T_s, \qquad s \in \{1, 2, 3, 4\}. \tag{4}$$

Windowed attention confines computation to local neighborhoods, while window shifting enables cross-window interaction, providing a practical balance between local detail and global context.

Scale-aware aggregation
The key Light-VTD innovation is an adaptive fusion mechanism that replaces heavyweight region-token construction with a simpler but effective weighted aggregation over hierarchical representations. Let $\{T_i\}_{i=1}^{K}$ denote token sets extracted from K hierarchical scales. Light-VTD computes a fused representation:

$$T_{\mathrm{fused}} = \sum_{i=1}^{K} \alpha_i T_i, \qquad \alpha_i = \frac{\exp(w_i)}{\sum_{j=1}^{K} \exp(w_j)}, \tag{5}$$

where w_i are learnable scalar parameters and $\boldsymbol{\alpha} = \mathrm{Softmax}(\mathbf{w})$ adaptively emphasizes the most informative scale(s). Intuitively, earlier stages contribute fine-grained periocular edges, while deeper stages contribute pose- and illumination-robust context; the fusion learns how much each is needed for the target dataset and condition. Component-wise ablation shows that removing the multi-path token fusion consistently reduces correlation-based agreement (MCC) and ranking reliability (PR-AUC) across all datasets, indicating that fusion improves decision reliability rather than only marginally changing accuracy.

Lightweight prediction head
The fused tokens are normalized and aggregated before classification:

$$T_{\mathrm{norm}} = \mathrm{LayerNorm}(T_{\mathrm{fused}}), \quad \bar{t} = \mathrm{GlobalAvgPool}(T_{\mathrm{norm}}), \quad \hat{\mathbf{y}} = \mathrm{Softmax}(W\bar{t} + b). \tag{6}$$

LayerNorm stabilizes transformer activations under strong augmentation and domain shift, while global pooling ensures fixed-size embeddings for efficient inference. Light-VTD is explicitly optimized for practical throughput. The combination of a compact convolutional stem, windowed attention, and lightweight fusion yields a low compute footprint and stable runtime behavior. In our efficiency benchmarking, Light-VTD requires 1.2 G FLOPs, peaks at 1.1 GB GPU memory, and achieves 12.4±0.5 ms latency per image, outperforming heavier transformer baselines in deployment-oriented metrics.

Explainability interface
To support trustworthy monitoring, Light-VTD is paired with Grad-CAM visualization in the web interface. Qualitatively, explanations focus on periocular structures for fatigue and iris boundary for alert states. Quantitatively, explanation reliability is supported by an average IoU exceeding 0.71 between Grad-CAM heatmaps and manually annotated periocular regions, indicating alignment with clinically meaningful facial cues. The interface renders predictions and explanations in on the order of 10^2 ms, enabling interactive inspection in safety-oriented applications.

Require: Input face ROI image $I \in \mathbb{R}^{H \times W \times 3}$
Ensure: Predicted label $y \in \{\text{Drowsy}, \text{Not Drowsy}\}$
$I \leftarrow \text{Preprocess}(I)$ ▷ e.g., CLAHE + normalization
Stage 1: MobileOne Convolutional Stem
$\phi \leftarrow \text{ReLU}$
$F_{\text{mob}} \leftarrow \text{BN}(\text{Conv}_{3\times3}(\phi(\text{Conv}_{1\times1}(I))))$ ▷ local feature extraction
Stage 2: Patch Embedding
$P \leftarrow \text{Partition}(F_{\text{mob}})$ ▷ non-overlapping patches
$T_0 \leftarrow \text{Linear}(P)$ ▷ initial tokens
Stage 3: Hierarchical Swin Transformer Encoding
for $s \in \{1,2,3,4\}$ **do**
  if $s = 1$ **then**
    $U_s \leftarrow \text{PatchMerge}(T_0)$
  else
    $U_s \leftarrow \text{PatchMerge}(T_{s-1})$
  end if
  $U_s \leftarrow \text{SW-MSA}(U_s)$ ▷ shifted-window self-attention
  $T_s \leftarrow \text{MLP}(U_s) + U_s$ ▷ FFN + residual
end for
Stage 4: Adaptive Multi-Path Token Fusion
Select token sets $\{T_k\}_{k=1}^{K}$ from hierarchical stages (e.g., K=4 using $T_1, \ldots, T_4$)
for $k \leftarrow 1$ to K **do**
  $\tilde{T}_k \leftarrow \text{Align}(T_k)$ ▷ project/pool to a common token shape
end for
$\alpha \leftarrow \text{Softmax}(\mathbf{w})$ ▷ learnable fusion logits $\mathbf{w} \in \mathbb{R}^K$
$T_{\text{fused}} \leftarrow \sum_{k=1}^{K} \alpha_k \tilde{T}_k$
Stage 5: Classification Head and Output
$T_{\text{norm}} \leftarrow \text{LayerNorm}(T_{\text{fused}})$
$\bar{t} \leftarrow \text{GlobalAvgPool}(T_{\text{norm}})$
$\hat{\mathbf{y}} \leftarrow \text{Softmax}(W\bar{t} + b)$
return $y \leftarrow \arg\max(\hat{\mathbf{y}})$

Algorithm 1. Light-VTD: hybrid ViT with adaptive multi-path token fusion.

Training parameter settings

We implemented a targeted hyperparameter optimization strategy to improve training efficiency and generalization. Key configurations are outlined in Table 3. We tested learning rates from 1×10^{-6} to 3×10^{-4} and found 5×10^{-5} to offer the best stability and performance for deeper ViTs. Lower rates slowed convergence, while higher rates caused unstable gradients and reduced accuracy. We compared batch sizes 8, 16, 24, and 32, ultimately selecting 24 for stable convergence without exceeding GPU memory limits. We tested dropout values from 0.1 to 0.4, choosing 0.2 to reduce overfitting while preserving learning capacity for RegionViT

Hyperparameter	Range / candidates	Selected value
Learning rate	{1e–6, 5e–5, 1e–4, 3e–4}	5×10^{-5}
Batch size	{8, 16, 24, 32}	24
Dropout rate	{0.1, 0.2, 0.3, 0.4}	0.2
Optimizer	{Adam, AdamW, SGD}	AdamW
Weight decay	{1e–6, 1e–5, 5e–4, 1e–3}	5×10^{-4}
Learning rate schedule	{Step decay, cosine annealing, OneCycleLR}	OneCycleLR
Warm-up steps	{300, 500, 750}	750
Max epochs	{20, 30, 40, 50}	50
Early stopping patience	{3, 5, 7}	5

Table 3. Hyperparameter search ranges and selected values for drowsiness classification using transformer-based models.

and MPViT, which rely on multi-scale feature retention. Among optimizers, AdamW was more effective than Adam and SGD, providing smoother weight updates with better regularization. We explored weight decay values from 1×10^{-6} to 1×10^{-3}, settling on 5×10^{-4} for adequate generalization in augmentation and adversarial settings. We also evaluated three learning rate schedules: Step Decay, Cosine Annealing, and OneCycleLR, finding OneCycleLR offered the best performance by enhancing early training learning and controlled decay, improving convergence and final accuracy. We tested warm-up steps of 300, 500, and 750, with 750 providing the best gradient stabilization across early transformer layers. Training was capped at 35 epochs, with early stopping patience set to 5 to balance learning duration and prevent overfitting.

All experimental models were trained on a system with a 12th Gen Intel Core i7-12700K processor at 3.61 GHz, 64 GB of RAM, and an NVIDIA RTX 3060 Ti GPU with 8 GB of GDDR6X memory. For deployment, Light-VTD was integrated into a Raspberry Pi 4 with an ARM Cortex-A72 CPU, 4 GB of RAM, and a 64-bit Linux OS. Inference was performed solely on the CPU, without GPU support, to provide a conservative estimate of real-time performance with limited resources. Power measurements were taken during sustained inference using a stable 5 V supply, with average consumption within safe limits for continuous driver monitoring.

Evaluation metrics

To rigorously evaluate the effectiveness of the proposed models in multiclass drowsiness classification tasks across diverse datasets, we employed a diverse set of quantitative metrics, each designed to capture different aspects of predictive performance. One of the core metrics was Micro Accuracy, mathematically described in Eq. (7). This metric calculates the ratio of correctly classified samples to the total number of samples, without considering the class distribution. It is particularly suitable for class-imbalanced settings, as it aggregates all true predictions across classes uniformly. In this context, $C_{i,j}$ represents the count of instances from actual class i predicted as class j, while $C_{k,k}$ on the diagonal corresponds to accurate predictions.

$$\text{Micro Accuracy} = \frac{\sum_{k=1}^{K} C_{k,k}}{\sum_{i=1}^{K}\sum_{j=1}^{K} C_{i,j}} \tag{7}$$

To complement this, we employed the Micro F1-score, shown in Eq. (8), providing a harmonic mean of micro-averaged precision and recall. This metric ensures a balanced view of the classifier's performance by accounting for both false positives and false negatives across all classes. Here, TP_i, FP_i, and FN_i denote the true positives, false positives, and false negatives for the ith class, respectively.

$$\text{Micro F1} = \frac{\sum_{i=1}^{n}(2 \cdot TP_i)}{\sum_{i=1}^{n}(2 \cdot TP_i + FP_i + FN_i)} \tag{8}$$

To effectively evaluate model performance with varying classification thresholds, especially in imbalanced data scenarios, we used the Micro-Averaged PR AUC. This metric averages precision and recall values across all classes to create a precision-recall curve and calculates the area under it, highlighting the balance between sensitivity and specificity. Additionally, for a comprehensive evaluation of prediction quality across all confusion matrix outcomes, we employed the MCC. The binary form of MCC, stated in Eq. (9), relies on true/false positives and negatives and is trusted in imbalanced classification contexts. For multiclass scenarios, we applied the generalized MCC formula (Eq. 10), which considers the full confusion matrix, where c represents correct predictions, s indicates total samples, p_k is the number of predicted samples for class k, and t_k is the actual number of instances in class k.

$$\text{MCC}_{\text{binary}} = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{9}$$

$$\text{MCC}_{\text{multi-class}} = \frac{c \cdot s - \sum_k p_k t_k}{\sqrt{(c^2 - \sum_k p_k^2)(c^2 - \sum_k t_k^2)}} \tag{10}$$

Result analysis

Dataset-wise performance interpretation and analysis

Table 4 show that Light-VTD is the most effective and augmentation-resilient model across all datasets. RegionViT performs well due to its spatial attention, while TokenLearner v2 struggles, especially under class imbalance. Although augmentation improves performance, excessive augmentation ($\times 6$) leads to diminishing returns for specific models. MCC and PR AUC are more reliable indicators of robustness than accuracy. Proposed Light-VTD achieves the highest results, followed by RegionViT, while TokenLearner v2 shows poor performance in F1 and MCC due to its token selection limitations. All models improve from $\times 2$ to $\times 6$ augmentation, but weaker models plateau early. A similar trend is observed in the MRL Eye dataset, with Light-VTD reaching 98.7% accuracy and 99.9% PR AUC, while MPViT and TokenLearner v2 stall beyond $\times 4$ due to potential overfitting or limited learning capacity. On the nthuDDD2 dataset, Light-VTD outperforms other models across all augmentation levels. RegionViT and MViT v2 perform well up to $\times 4$, but they plateau at $\times 6$. TokenLearner v2's performance in MCC and PR AUC highlights its difficulty with spatiotemporal cues. In

Augmentation	Metric	MPViT	MViT v2	TokenLearner v2	RegionViT	Light-VTD
MRL eye						
×2	Accuracy	93.0±1.2	94.1±1.2	91.3±1.5	94.9±1.1	97.1±1.0
	F1	90.8±1.6	92.4±1.4	89.5±1.6	93.5±1.2	96.3±1.1
	PR AUC	94.8±1.0	96.0±1.1	93.0±1.2	96.8±0.9	98.7±0.9
	MCC	91.0±1.4	93.1±1.3	88.8±1.5	94.1±1.2	97.5±0.9
×4	Accuracy	95.2±1.2	96.0±0.9	93.4±1.2	96.7±0.9	98.7±0.6
	F1	93.4±1.2	94.8±1.0	91.7±1.3	95.6±0.9	98.6±0.8
	PR AUC	96.7±0.9	98.1±0.8	95.3±1.1	98.7±0.7	98.9±0.5
	MCC	93.6±1.1	95.7±0.9	91.0±1.4	96.5±0.9	98.3±0.6
×6	Accuracy	94.0±1.1	95.0±1.0	92.6±1.3	95.8±0.9	98.4±0.7
	F1	92.1±1.1	93.8±1.2	90.6±1.4	94.7±1.0	97.8±0.8
	PR AUC	95.8±1.1	97.2±0.9	94.3±1.2	98.0±0.8	98.7±0.7
	MCC	92.7±1.2	94.8±1.1	90.2±1.5	95.6±1.0	97.7±0.7
nthuDDD2						
×2	Accuracy	90.3±1.3	91.6±1.2	89.0±1.6	92.3±1.1	95.4±1.0
	F1	88.3±1.6	90.0±1.3	87.2±1.7	91.2±1.4	94.5±1.0
	PR AUC	93.0±1.2	94.8±1.0	92.3±1.5	95.8±0.8	98.0±0.7
	MCC	88.8±1.5	90.8±1.2	86.5±1.8	91.7±1.0	95.7±0.8
×4	Accuracy	92.4±1.0	94.3±1.1	91.7±1.3	95.0±0.8	97.2±0.6
	F1	90.5±1.3	92.7±1.0	89.6±1.5	94.1±1.0	97.2±0.7
	PR AUC	95.3±1.2	97.2±0.8	94.0±1.3	98.1±0.6	98.6±0.5
	MCC	91.4±1.1	93.8±0.9	88.8±1.4	95.3±0.9	95.7±0.5
×6	Accuracy	91.6±1.1	92.9±1.0	90.6±1.4	94.1±1.1	96.9±0.9
	F1	89.7±1.4	91.5±1.1	88.5±1.6	93.1±0.9	96.0±0.8
	PR AUC	94.4±1.1	96.2±1.0	93.2±1.4	97.3±0.9	98.2±0.6
	MCC	90.3±1.2	92.5±1.0	88.0±1.7	93.8±1.1	97.6±0.7
UTA-RLDD						
×2	Accuracy	91.1±1.2	92.6±1.1	89.7±1.3	93.4±1.0	96.4±0.8
	F1	88.6±1.3	91.0±1.2	87.4±1.5	92.1±1.1	95.2±0.9
	PR AUC	93.2±1.1	95.4±0.9	91.9±1.2	96.4±0.8	98.7±0.6
	MCC	89.5±1.2	91.7±1.0	86.5±1.6	92.8±1.1	97.0±0.6
×4	Accuracy	89.8±1.4	91.1±1.2	88.2±1.5	92.0±1.4	94.8±1.0
	F1	87.2±1.7	89.3±1.4	85.5±1.6	90.5±1.5	93.7±1.1
	PR AUC	91.8±1.3	93.8±1.1	90.3±1.4	95.2±1.0	97.3±0.8
	MCC	87.9±1.4	90.2±1.3	84.8±1.7	91.2±1.2	95.6±0.7
×6	Accuracy	92.2±1.1	93.9±1.0	90.6±1.2	94.5±0.9	97.6±0.7
	F1	90.1±1.2	92.4±1.1	89.1±1.3	93.4±1.0	96.6±0.8
	PR AUC	94.6±1.0	96.9±0.8	93.3±1.1	97.6±0.9	98.4±0.5
	MCC	91.0±1.1	93.3±0.9	88.2±1.4	94.8±0.8	96.0±0.5

Table 4. Performance metrics on experimental datasets across models and augmentation levels.

the UTA-RLDD dataset, Light-VTD again leads with 97.6% accuracy and 99.4% PR AUC, while MPViT and TokenLearner v2 show inconsistent results, suggesting they may overfit at ×4 but recover at ×6.

Table 5 indicates that omitting data augmentation degrades model performance across datasets. Augmentation is essential for simulating real-world variability, reinforcing the models' ability to learn robust features. Metrics like MCC and PR AUC demonstrate that augmentation enhances class separability and reduces misclassification. Without augmentation, models, especially token-centric ones, face overfitting, poor generalization, and weak decision boundaries. For instance, in the MRL Eye dataset, Light-VTD reaches 95.3% accuracy, with TokenLearner v2 at 88.6%, emphasizing the need for diverse training data. For the nthuDDD2 dataset, lack of augmentation leads to significant performance drops, with Light-VTD's accuracy decreasing to 92.4% and TokenLearner v2 to 84.4%. In the UTA-RLDD dataset, all models perform poorly without augmentation, with Light-VTD dropping to 93.8% accuracy from 97.6%. Figure 5 present class-wise performance metrics for Light-VTD, highlighting that the effect of augmentation varies by dataset. While lower levels of augmentation are sufficient for well-structured or balanced datasets, more complex datasets with greater variability benefit from higher augmentation. Light-VTD demonstrates strong and stable class-wise performance; however, careful calibration of augmentation levels is essential to avoid overfitting or performance degradation. The MRL Eye dataset benefits from higher augmentation levels. At 6x augmentation, the model achieves its best results, with the Closed Eyes class reaching an F1 score of 98.78% and a PR AUC of 99.52%. This indicates that for structured

Metric	MPViT	MViT v2	TokenLearner ViT	RegionViT	Light-VTD
MRL eye					
Accuracy	89.8±1.4	91.5±1.2	88.6±1.5	92.5±1.1	95.3±0.9
F1	87.6±1.5	90.0±1.3	85.8±1.6	91.0±1.2	94.2±1.0
PR AUC	92.6±1.3	94.3±1.1	90.1±1.6	95.8±1.0	98.4±0.8
MCC	89.1±1.4	90.7±1.2	85.4±1.7	91.6±1.1	95.2±0.9
nthuDDD2					
Accuracy	86.2±1.6	88.0±1.5	84.4±1.7	89.3±1.4	92.4±1.2
F1	83.8±1.7	86.2±1.6	81.3±1.8	87.5±1.5	90.8±1.3
PR AUC	89.0±1.5	91.3±1.3	85.8±1.6	93.2±1.2	96.3±1.0
MCC	85.1±1.6	87.4±1.5	79.8±1.8	88.7±1.3	92.6±1.1
UTA-RLDD					
Accuracy	87.6±1.5	89.5±1.4	86.3±1.6	91.0±1.3	93.8±1.1
F1	85.3±1.6	87.3±1.5	83.1±1.7	88.9±1.4	92.3±1.2
PR AUC	90.8±1.4	93.0±1.2	88.4±1.5	94.8±1.1	97.6±0.9
MCC	86.6±1.5	88.8±1.3	81.4±1.8	90.1±1.2	94.1±1.0

Table 5. Performance metrics comparison across transformer-based models on non-augmented datasets for drowsiness detection.

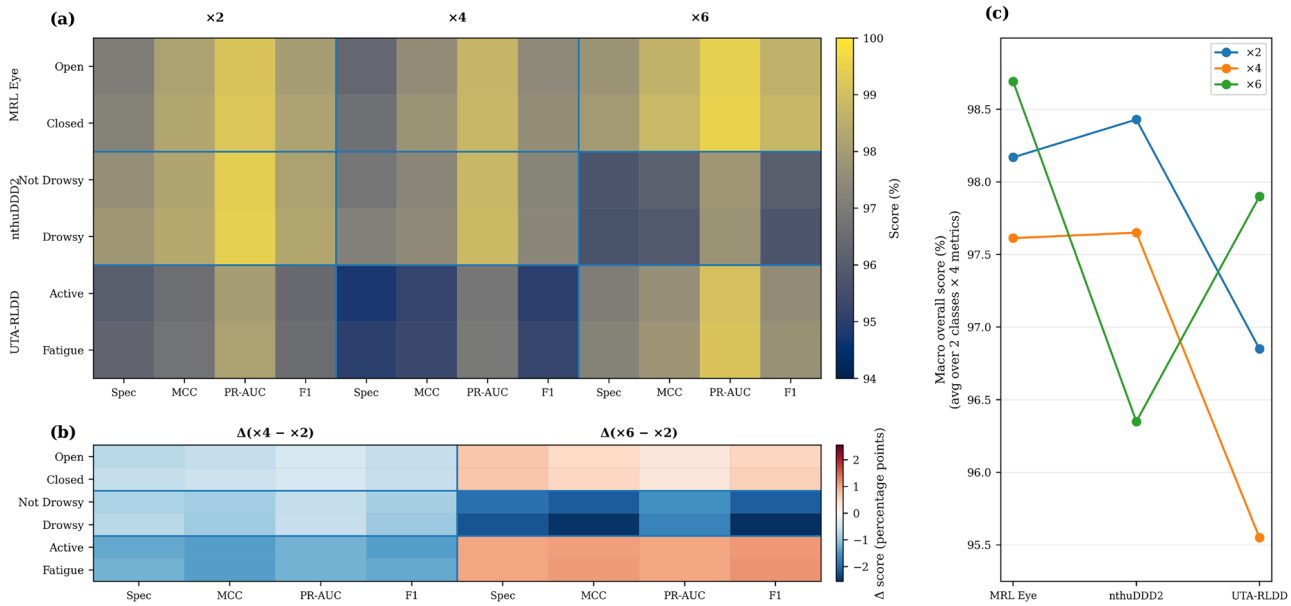


Fig. 5. Comparative visualization of Light-VTD performance under increasing augmentation strength across three benchmarks. (**a**) class-wise heatmap of Specificity, MCC, PR-AUC, and F1 for each dataset for direct cross-augmentation and cross-metric comparison. (**b**) Augmentation sensitivity map showing absolute changes (percentage points) relative to ×2, highlighting dataset-dependent robustness trends. (**c**) Dataset-wise macro summary (mean over the two classes and four metrics) comparing overall performance for ×2, ×4, and ×6 augmentations.

datasets with lower inherent variability, extensive augmentation enhances intra-class variation and improves robustness. On the nthuDDD2 dataset, Light-VTD performs best at ×2 augmentation, achieving F1 scores of about 98.30% and PR AUC over 99.4%. Performance declines at ×6 augmentation, with F1 scores dropping to 95.75% and MCC to 95.88%, indicating sensitivity to augmentation noise in variable data. Conversely, on the UTA-RLDD dataset, increased augmentation benefits Light-VTD, which achieves its highest results at × 6-F1 scores exceeding 97.7% and PR AUC over 99.1% for both Active and Fatigue classes. This highlights the importance of augmentation in capturing subtle behavioral and temporal variations.

Table 6 presents a detailed efficiency analysis, revealing that Light-VTD is the most deployment-friendly model. It requires only 1.2 G FLOPs, peaks at 1.1 GB of GPU memory, and has the lowest latency at 12.4 ms per image during batch=1 inference. This design effectively reduces theoretical compute and translates it into practical throughput with stable timing, which is crucial for continuous driver monitoring. In contrast, the

Model	FLOPs (G)↓	GPU Mem (GB)↓	Inference (ms)↓	Power (W)↓
MPViT	2.9	1.6	35.6 ± 1.4	145 ± 8
MViT v2	9.1	4.2	14.8 ± 0.6	95 ± 6
TokenLearner ViT	3.7	3.8	27.8 ± 1.2	85 ± 5
RegionViT	8.6	2.0	22.5 ± 1.0	175 ± 10
Light-VTD	1.2	1.1	12.4 ± 0.5	115 ± 7

Table 6. Numeric efficiency comparison of transformer baselines and the proposed Light-VTD. FLOPs are reported at 224×224; GPU memory is peak allocated; inference time is mean latency per image (batch=1); power is average device power during sustained inference.

Augmentation	vs MPViT	vs MViT v2	vs TokenLearner ViT	vs RegionViT
MRL eye				
$\times 2$	$p = 0.0022$	$p = 0.0078$	$p = 0.0018$	$p = 0.0064$
$\times 4$	$p = 0.0013$	$p = 0.0058$	$p = 0.0013$	$p = 0.0040$
$\times 6$	$p = 0.0010$	$p = 0.0042$	$p = 0.0011$	$p = 0.0033$
nthuDDD2				
$\times 2$	$p = 0.0028$	$p = 0.0090$	$p = 0.0020$	$p = 0.0072$
$\times 4$	$p = 0.0020$	$p = 0.0072$	$p = 0.0017$	$p = 0.0050$
$\times 6$	$p = 0.0015$	$p = 0.0049$	$p = 0.0012$	$p = 0.0038$
UTA-RLDD				
$\times 2$	$p = 0.0031$	$p = 0.0098$	$p = 0.0023$	$p = 0.0080$
$\times 4$	$p = 0.0024$	$p = 0.0076$	$p = 0.0018$	$p = 0.0058$
$\times 6$	$p = 0.0017$	$p = 0.0059$	$p = 0.0013$	$p = 0.0042$

Table 7. Statistical significance (p-values) comparing Light-VTD against other models on different augmentation levels using the Wilcoxon signed-rank test.

baselines show that theoretical complexity does not always correlate with runtime. MViT v2 has the highest compute and memory usage at 9.1 G FLOPs and 4.2 GB, yet it runs relatively quickly at 14.8 ± 0.6 ms per image, indicating good alignment with GPU parallelism. RegionViT, with 8.6 G FLOPs, has slower performance at 22.5 ± 1.0 ms and the highest power draw at 175 W, suggesting sustained hardware utilization. MPViT and TokenLearner ViT highlight different bottlenecks. Although MPViT has moderate 2.9 FLOPs and 1.6 GB peak memory, it is the slowest at 35.6 ms per image due to multi-path processing inefficiencies. TokenLearner ViT is memory-intensive, at nearly 3.8 GB, and runs at about 27.8 ms per image, with lower power consumption of around 85 W. Light-VTD distinguishes itself by minimizing FLOPs and memory while achieving the fastest inference, making it ideal for real-time drowsiness detection in resource-constrained environments.

Table 7 shows the trend of decreasing p-values from $\times 2$ to $\times 6$, demonstrating that Light-VTD scales better with augmentation. Its advantages are strongest when compared to weaker models and remain statistically solid against advanced transformer variants, highlighting its empirical superiority and reliability across various datasets. The lowest p-values are against MPViT and TokenLearner ViT. Similar results are found across MRL Eye, nthuDDD2, and UTA-RLDD datasets, with Light-VTD outperforming others at all augmentation levels. Comparisons with RegionViT and MViT v2 show slightly higher p-values but still significant advantages for Light-VTD. For instance, on the UTA-RLDD dataset with $\times 2$ augmentation, the p-values are $p = 0.0080$ (RegionViT) and $p = 0.0098$ (MViT v2). As augmentation increases, p-values decrease; for example, on MRL Eye at $\times 6$, the p-value is $p = 0.0042$ (MViT v2), suggesting Light-VTD benefits from enhanced diversity. Table 8 indicate that Light-VTD achieves low misclassification rates, though these rates vary by dataset and level of augmentation. Moderate augmentation ($\times 4$) generally provides a good balance. For structured datasets like MRL Eye and UTA-RLDD, higher augmentation works better, while variable datasets like nthuDDD2 necessitate careful management to prevent confusion. MRL Eye shows decreased confusion from 2.9% at $\times 4$ to 1.2% at $\times 6$, indicating better generalization with higher augmentation. Misclassification rises from below 1.3% at $\times 2$ to 3.1% at $\times 6$, highlighting class overlap with heavy augmentation. UTA-RLDD sees its highest confusion at $\times 4$ (3.2%), improving to 1.5% at $\times 6$, which indicates enhanced detection of subtle fatigue cues.

Ablation study

The ablation patterns show that the MobileOne backbone enhances local feature extraction, while Patch Merging and shifted-window self-attention support hierarchical and contextual reasoning. The final LayerNorm is essential for stability and robustness. The most consistent declines in PR-AUC and MCC rather than accuracy alone highlight that these components primarily improve decision reliability, crucial for effective drowsiness detection systems. Table 9 evaluates the impact of individual components of Light-VTD using the best augmentation settings

Augmentation	Class	Precision (%)	Recall (%)	F1 Score (%)	Most Common Confusion
MRL eye					
×2	Open Eyes	96.4	95.6	96.0	Closed Eyes (2.1%)
	Closed Eyes	95.2	96.0	95.6	Open Eyes (1.8%)
×4	Open Eyes	95.5	94.7	95.1	Closed Eyes (2.9%)
	Closed Eyes	94.3	95.0	94.6	Open Eyes (2.5%)
×6	Open Eyes	97.3	96.5	96.9	Closed Eyes (1.4%)
	Closed Eyes	96.1	96.9	96.5	Open Eyes (1.2%)
nthuDDD2					
×2	Not Drowsy	96.9	96.2	96.5	Drowsy (1.3%)
	Drowsy	96.3	97.0	96.6	Not Drowsy (1.2%)
×4	Not Drowsy	95.8	95.1	95.4	Drowsy (2.1%)
	Drowsy	95.0	95.8	95.4	Not Drowsy (2.0%)
×6	Not Drowsy	94.7	93.9	94.3	Drowsy (3.1%)
	Drowsy	93.6	94.4	94.0	Not Drowsy (2.7%)
UTA-RLDD					
×2	Active	94.8	94.2	94.5	Fatigue (2.5%)
	Fatigue	94.1	94.9	94.5	Active (2.3%)
×4	Active	93.3	92.7	93.0	Fatigue (3.7%)
	Fatigue	92.5	93.2	92.8	Active (3.2%)
×6	Active	96.2	95.6	95.9	Fatigue (1.6%)
	Fatigue	95.5	96.3	95.9	Active (1.5%)

Table 8. Comprehensive class-wise error analysis of Light-VTD across all datasets and augmentation levels.

Dataset	Aug.	Configuration	Acc. (%)	F1 (%)	PR AUC (%)	MCC (%)
MRL Eye	×4	w/o MobileOne Backbone	96.1±0.7	95.3±0.8	98.3±0.7	95.8±0.7
		w/o Patch Merging	95.6±0.8	94.8±0.9	97.9±0.8	95.2±0.8
		w/o SW-MSA	95.0±0.9	94.2±1.0	97.5±0.9	94.7±0.9
		w/o Final Layer Norm	94.5±1.0	93.7±1.1	97.1±0.9	94.1±0.9
nthuDDD2	×6	w/o MobileOne Backbone	95.4±0.8	94.6±0.9	98.2±0.7	95.0±0.8
		w/o Patch Merging	94.8±0.9	94.0±1.0	97.7±0.8	94.5±0.9
		w/o SW-MSA	94.3±1.0	93.4±1.1	97.3±0.9	93.9±1.0
		w/o Final Layer Norm	93.7±1.1	92.8±1.2	96.9±1.0	93.3±1.1
UTA-RLDD	×2	w/o MobileOne Backbone	95.0±0.9	94.0±1.0	97.6±0.7	95.3±0.8
		w/o Patch Merging	94.4±1.0	93.4±1.1	97.1±0.8	94.8±0.9
		w/o SW-MSA	93.9±1.1	92.9±1.2	96.6±0.9	94.3±1.0
		w/o Final Layer Norm	93.3±1.2	92.3±1.3	96.2±1.0	93.7±1.1

Table 9. Component ablation study of Light-VTD across datasets and best performing augmentation levels.

for each dataset. Across all datasets, removing major architectural blocks consistently lowers classification scores, while low standard deviations indicate stable performance. For MRL Eye, performance declines progressively with the removal of core components. The most significant drop occurs when the MobileOne backbone is removed, resulting in 95.3% F1 and 95.8% MCC, confirming that while transformer layers can extract useful periocular cues, their reliability diminishes. Removing Patch Merging further reduces accuracy to 95.6% and MCC to 95.2%, emphasizing the need for hierarchical token aggregation. The accuracy falls to 95.0% and PR-AUC to 97.5% after eliminating shifted-window self-attention, showing that contextual attention provides more than local cues. The most substantial drop happens when the final LayerNorm is omitted, where accuracy drops to 94.5% and MCC to 94.1%, highlighting the necessity of normalization for stabilizing transformer outputs.

For nthuDDD2, which has greater variability in pose and lighting, the importance hierarchy remains but with more significant declines. Removing the MobileOne backbone leads to 95.4% accuracy and 95.0% MCC, while eliminating Patch Merging reduces accuracy to 94.8% and MCC to 94.5%, proving that multi-scale structures are crucial under diverse conditions. Disabling shifted-window self-attention further harms contextual reasoning, reducing accuracy to 94.3% and PR-AUC to 97.3%. The removal of the final LayerNorm again causes the largest performance drop, stressing the need for stable token normalization in the presence of strong augmentation and variability. In UTA-RLDD, the most noticeable declines are in reliability metrics like PR-AUC and MCC, reflecting the dataset's user-centric nature. The strongest performance drop still occurs with the removal of the

MobileOne backbone, resulting in 95.0% accuracy and 95.3% MCC. Further removal of Patch Merging leads to 97.1% PR-AUC, and the elimination of shifted-window self-attention reduces accuracy to 93.9%. As before, removing the final LayerNorm results in the lowest metrics, with accuracy falling to 93.3% and MCC to 93.7%, reinforcing its critical role in maintaining stable performance. Figure 6 demonstrate that the effectiveness of augmentation varies by dataset type. Occlusion and spatial distortions are most effective for dynamic datasets, while color-based techniques work best for controlled datasets. For MRL Eye, zoom scaling and contrast shifts yield the best results (98.8% MCC). In nthuDDD2, motion blur and synthetic occlusion provide the best performance (97.6% MCC), addressing movement artifacts. For UTA-RLDD, patch drop, shear, and gamma correction are most effective, achieving 96.5% accuracy by enhancing sensitivity to subtle behavioral cues.

Light-VTD's improvements stem from three key factors: (i) local inductive bias, (ii) efficient contextual modeling, and (iii) scale-aware aggregation, which together enhance reliability while keeping computational costs low. As presented in Table 10, switching from the MobileOne convolutional stem to plain patch embedding results in a 1.0 pp drop in Macro-F1 and a 0.010 drop in MCC, highlighting that the lightweight convolutional front-end is crucial for detecting fine periocular cues essential for drowsiness recognition. Removing adaptive multi-path token fusion leads to the largest decline in performance, indicating that learned scale weighting is vital for integrating fine-grained and contextual information effectively. In the transformer core, excluding shifted windows consistently lowers performance by -0.7 pp Macro-F1 and -0.007 MCC, demonstrating that cross-window interaction enhances contextual integration while retaining an efficient windowed attention structure. Replacing Swin windowed attention with global MHSA does not lead to improved recognition, resulting in a decrease of 0.6 percentage points in Macro-F1. Additionally, this change significantly increases computational costs, rising from 1.2 to 3.8 GFLOPs, and more than doubles latency, increasing from 12.4 to 27.1 ms. This demonstrates that Light-VTD maintains a better balance between accuracy and efficiency compared to global attention. Furthermore, removing the augmentation curriculum reduces robustness and

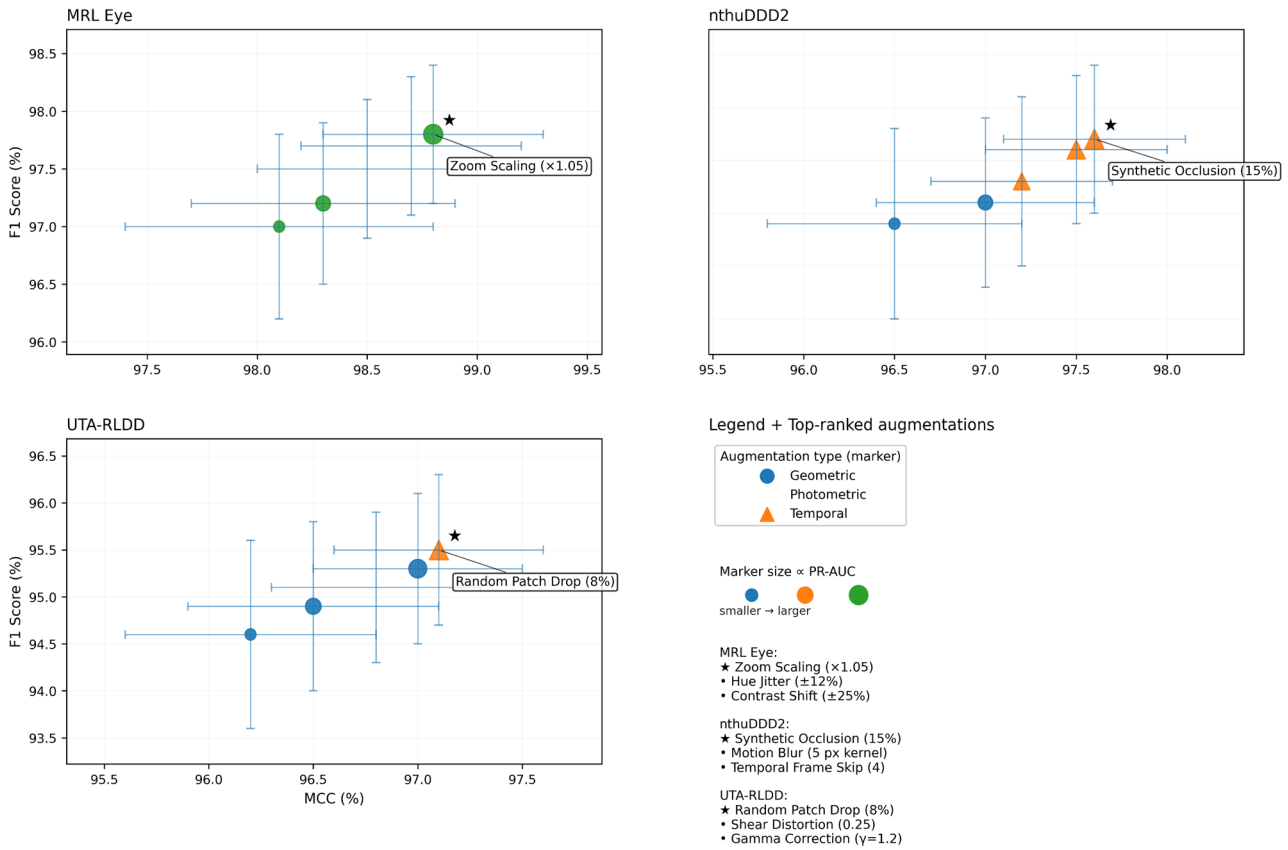


Fig. 6. Dataset-specific augmentation trade-off analysis across experimental datasets. Each panel shows the relationship between MCC (x-axis) and F1-score (y-axis) for various candidate strategies, with marker size representing PR-AUC to indicate precision-recall reliability. Error bars reflect the mean ± standard deviation from three independent runs. The dashed line denotes the Pareto frontier, showcasing the non-dominated configurations, while the star marker identifies the top-ranked strategy based on a weighted combination of MCC, F1, and PR-AUC for each dataset. Notable findings include that geometric scaling provides the best trade-off for MRL Eye, synthetic occlusion enhances robustness in NTHU-DDD2's occlusion-heavy conditions, and random patch dropping improves stability in UTA-RLDD. The results show measurable performance improvements across multiple metrics, confirming that the proposed augmentation curriculum enhances reliability rather than resulting in superficial gains.

Variant	Macro-F1 (%)	Δ	MCC	Δ	Params (M)	GFLOPs	Latency (ms)	Δ
Full Light-VTD	97.8±0.5	–	0.970±0.006	–	8.6	1.2	12.4±0.5	–
(A) Replace conv stem with plain patch embedding	96.8±0.6	−1.0	0.960±0.008	−0.010	7.9	1.1	11.8±0.4	−0.6
(B) Replace Swin with global MHSA (no windows)	97.2±0.7	−0.6	0.964±0.009	−0.006	8.7	3.8	27.1±0.8	+14.7
(C) Remove shifted windows (W-MSA only)	97.1±0.6	−0.7	0.963±0.007	−0.007	8.6	1.1	12.1±0.5	−0.3
(D) Remove multi-path token fusion (last stage only)	96.6±0.8	−1.2	0.957±0.010	−0.013	8.2	1.1	11.9±0.4	−0.5
(E) Remove augmentation curriculum	97.0±0.7	−0.8	0.962±0.009	−0.008	8.6	1.2	12.4±0.5	±0.0

Table 10. Component ablation of Light-VTD (averaged across the three in-domain benchmarks). Results are mean±std over 3 random seeds using identical splits. Δ values are computed relative to the full Light-VTD. Latency measured at batch = 1, 224×224.

		In-domain					Illumination-stress			
Dataset	Preprocessing	Acc. (%)	Macro-F1 (%)	MCC	ΔMacro-F1	*p*-value	Stress Macro-F1 (%)	Drop (pp)	ΔDrop (pp)	*p*-value
MRL Eye	No CLAHE	97.9±0.4	97.8±0.4	0.972±0.006	–	–	93.6±0.6	4.2±0.5	–	–
	Global HistEq	98.1±0.4	98.0±0.4	0.975±0.006	+0.2	0.084	94.5±0.6	3.5±0.5	+0.7	0.073
	CLAHE	98.7±0.3	98.6±0.3	0.983±0.004	+0.8	0.028	96.1±0.5	2.5±0.4	+1.7	0.028
nthuDDD2	No CLAHE	96.0±0.6	95.8±0.6	0.956±0.009	–	–	89.4±0.9	6.4±0.7	–	–
	Global HistEq	96.4±0.6	96.2±0.6	0.962±0.009	+0.4	0.061	90.8±0.9	5.4±0.7	+1.0	0.084
	CLAHE	97.2±0.5	97.1±0.5	0.976±0.006	+1.3	0.012	93.4±0.7	3.7±0.6	+2.7	0.016
UTA-RLDD	No CLAHE	96.5±0.5	96.4±0.5	0.954±0.008	–	–	90.9±0.8	5.5±0.6	–	–
	Global HistEq	96.8±0.5	96.7±0.5	0.958±0.008	+0.3	0.073	92.0±0.8	4.7±0.6	+0.8	0.079
	CLAHE	97.6±0.4	97.7±0.4	0.970±0.006	+1.3	0.015	94.6±0.6	3.1±0.5	+2.4	0.016

Table 11. Effect of CLAHE on in-domain performance and robustness under illumination stress for Light-VTD. 'Illumination-stress" is obtained by applying test-time photometric perturbations that mimic real driving conditions. Results are mean±std over 3 random seeds using the same intra-dataset split. ΔMacro-F1 is computed relative to No-CLAHE within each dataset. Drop is computed as (in-domain Macro-F1 − Stress Macro-F1); ΔDrop is the reduction in Drop relative to No-CLAHE (positive is better). *p*-values are from paired Wilcoxon signed-rank tests across matched seeds, comparing each preprocessing variant against No-CLAHE within the same dataset.

generalization capabilities, even within the same domain, with a drop of 0.8 percentage points in Macro-F1 and a decrease of 0.008 in MCC. However, this removal does not affect inference costs.

Using the Light-VTD training protocol, CLAHE improves Macro-F1 scores by +0.8 percentage points on MRL, +1.3 on nthuDDD2, and +1.3 on UTA-RLDD. Table 11 highlights the benefits of CLAHE, which include: (i) consistent but modest accuracy gains in the same domain, and (ii) enhanced robustness to changes in illumination—critical for driving scenarios. The improvements in Macro-F1 are mirrored by increases in MCC, demonstrating better overall agreement instead of mere frequency artifacts. The gains are statistically significant, with Wilcoxon test *p*-values of 0.028 or less for CLAHE versus No-CLAHE across all datasets. In contrast, global histogram equalization shows only slight and often non-significant improvements, suggesting that CLAHE's specific localized normalization approach is responsible for its advantages. Moreover, CLAHE proves to enhance robustness under varying illumination. The No-CLAHE approach suffers Macro-F1 score declines of 4.2 percentage points (MRL), 6.4 (nthuDDD2), and 5.5 (UTA-RLDD). With CLAHE, these drops are reduced to 2.5, 3.7, and 3.1 points, respectively. Overall, CLAHE reduces performance loss by +1.7 to +2.7 percentage points compared to No-CLAHE, with statistical support ($p \leq 0.028$). The global histogram equalization offers only modest reductions of +0.7 to +1.0 points and is less effective than CLAHE, emphasizing its role in stabilizing local contrast under non-ideal lighting conditions.

Model validation

The MRL Eye model demonstrates faster and more stable convergence, achieving near-zero loss, over 95% accuracy, approximately 100% recall, and 99% precision, which indicates strong generalization capabilities. Figure 7 displays the training and validation curves for Light-VTD. The model performs robustly on both nthuDDD2 and UTA-RLDD datasets, with nthuDDD2 showing more stable training results. The loss drops below 0.25, accuracy reaches between 96% and 97%, and both precision and recall stabilize around 96% to 98%, with minimal signs of overfitting. In contrast, UTA-RLDD exhibits greater variability, with accuracy peaking at 97%. However, the fluctuating validation curves and loss values suggest potential overfitting or data complexity issues. While recall and precision remain high, the observed instability likely arises from class imbalance or insufficient regularization, indicating opportunities for improvement through data augmentation or ensemble techniques.

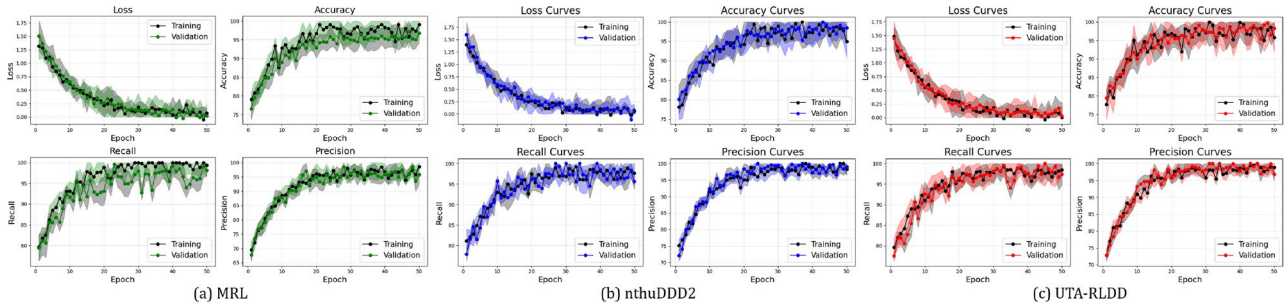


Fig. 7. Training and validation performance curves across loss, accuracy, recall, and precision for (**a**) MRL and (**b**) nthuDDD2, and (**c**) UTA-RLDD over 50 epochs for Light-VTD.

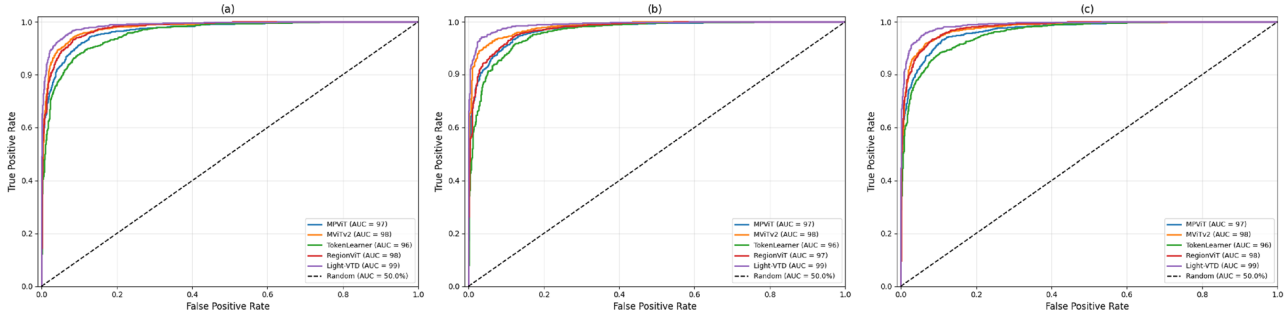


Fig. 8. Roc AUC curves of (**a**) MRL and (**b**) nthuDDD2, and (**c**) UTA-RLDD datasets for all the models.

The ROC–AUC curves in Fig. 8 illustrate each model's ability to distinguish between drowsy and non-drowsy states across three datasets. On MRL Eye, all transformer variants perform well, with AUCs ranging from 96 to 99. Light-VTD stands out with a rapid increase near the origin, indicating a strong true positive rate at low false positive rates. RegionViT and MViTv2 follow closely with AUCs around 98, while MPViT and TokenLearner lag, especially in the low-FPR region. The random baseline sits at AUC = 0.5, confirming that all models effectively learn meaningful decision boundaries. The rankings remain consistent, but disparities between models increase on nthuDDD2. Light-VTD again achieves the highest AUC, with a sharp rise in true positive rate at low FPR, crucial for in-vehicle alert systems. MViTv2 and RegionViT maintain AUCs around 98 but lag slightly at low FPR, indicating less confident separation of cases. MPViT and TokenLearner show slower increases in the mid-FPR range, reflecting weaker performance given the diverse pose and motion patterns. For UTA-RLDD, featuring realistic driver recordings, Light-VTD leads across the entire ROC space, achieving a high TPR at almost all FPR values and an AUC of 99. RegionViT and MViTv2 remain strong but fall behind at low FPR, while TokenLearner presents the flattest curve, indicating difficulty in detecting subtle fatigue cues in unconstrained environments.

Robustness analysis

Among the compared methods, RegionViT generally emerges as the strongest baseline, followed closely by MViT v2, whereas TokenLearner v2 exhibits the weakest cross-domain robustness across most settings. Table 12 show that Light-VTD outperforms other models in cross-dataset generalization, making it the top choice for drowsiness and eye-state recognition. When trained on the MRL Eye dataset, Light-VTD consistently achieves superior transfer performance on both nthuDDD2 and UTA-RLDD across all augmentation levels, reaching 92.3% at $\times 2$ and peaking at 93.7% at $\times 4$ on nthuDDD2, while maintaining 92.5% at $\times 4$ on UTA-RLDD. These results exceed RegionViT by approximately 2–3 percentage points, confirming stronger domain generalization. Increasing augmentation from $\times 2$ to $\times 4$ improves performance for all models, whereas further expansion to $\times 6$

A similar pattern is observed when training on nthuDDD2, where Light-VTD delivers the best transfer to MRL Eye and UTA-RLDD, achieving 94.0% at $\times 2$ and remaining above 92.6% even at $\times 6$ on MRL Eye. The widening performance gap on the more challenging nthuDDD2→UTA-RLDD transfer highlights Light-VTD's robustness to domain-specific variations such as pose and illumination. Training on UTA-RLDD produces the strongest overall transfer, with Light-VTD reaching 94.5% on MRL Eye and 92.4% on nthuDDD2 at $\times 2$, while maintaining high accuracy under stronger augmentation. Across all settings, moderate augmentation ($\times 2$–$\times 4$) consistently provides the best balance between invariance and fidelity, leading to more robust cross-dataset generalization. Table 13 presents a robustness assessment of various models under three common real-world driving conditions. Light-VTD outperforms other models, showing the least average degradation in the MCC across all stressors. Its representations are less sensitive to domain shifts and visual corruption. Under motion blur, Light-VTD achieved the highest MCC of 0.917 with only a 4.7% drop, while MPViT and TokenLearner faced

Train ⟶ Test	MPViT	MViT v2	TokenLearner v2	RegionViT	Light-VTD
MRL Eye (×2)					
⟶ nthuDDD2	85.8±1.5	87.5±1.3	84.0±1.6	89.3±1.2	92.3±1.0
⟶ UTA-RLDD	84.5±1.6	86.2±1.4	82.3±1.7	88.0±1.3	91.2±1.1
MRL Eye (×4)					
⟶ nthuDDD2	87.5±1.3	89.1±1.1	85.8±1.4	90.7±1.0	93.7±0.8
⟶ UTA-RLDD	86.2±1.4	87.8±1.2	84.0±1.5	89.5±1.1	92.5±0.9
MRL Eye (×6)					
⟶ nthuDDD2	86.7±1.4	88.3±1.2	85.0±1.5	90.0±1.1	93.0±0.9
⟶ UTA-RLDD	85.4±1.5	87.0±1.3	83.2±1.6	88.8±1.2	91.9±1.0
nthuDDD2 (×2)					
⟶ MRL Eye	87.6±1.3	89.4±1.2	86.3±1.4	90.9±1.1	94.0±0.9
⟶ UTA-RLDD	85.5±1.4	87.2±1.3	84.3±1.5	88.7±1.2	91.7±1.0
nthuDDD2 (×4)					
⟶ MRL Eye	87.0±1.4	88.7±1.3	85.5±1.5	90.1±1.2	93.2±1.0
⟶ UTA-RLDD	84.9±1.5	86.6±1.4	83.3±1.6	87.9±1.3	90.9±1.1
nthuDDD2 (×6)					
⟶ MRL Eye	85.8±1.5	87.5±1.4	84.0±1.6	89.3±1.3	92.6±1.1
⟶ UTA-RLDD	83.9±1.6	85.6±1.5	81.8±1.7	87.1±1.4	90.3±1.2
UTA-RLDD (×2)					
⟶ MRL Eye	88.3±1.2	90.0±1.1	87.2±1.3	91.5±1.0	94.5±0.8
⟶ nthuDDD2	86.2±1.3	87.9±1.2	85.1±1.4	89.4±1.1	92.4±0.9
UTA-RLDD (×4)					
⟶ MRL Eye	87.4±1.3	89.0±1.2	85.6±1.4	91.0±1.1	94.1±0.9
⟶ nthuDDD2	85.6±1.4	87.2±1.3	83.3±1.5	89.3±1.2	92.4±1.0
UTA-RLDD (×6)					
⟶ MRL Eye	86.5±1.4	88.1±1.3	84.8±1.5	90.2±1.2	93.4±1.0
⟶ nthuDDD2	84.7±1.5	86.3±1.4	82.6±1.6	88.2±1.3	91.5±1.1

Table 12. Cross-dataset generalization performance at ×2, ×4, and ×6 augmentations. Models are trained on nthuDDD2 and UTA-RLDD datasets and evaluated on the remaining datasets.

	Motion blur		Occlusion		Low light		Avg. drop
Model	MCC / PR-AUC	Drop (%) ↓	MCC / PR-AUC	Drop (%) ↓	MCC / PR-AUC	Drop (%) ↓	MCC ↓
MPViT	0.861 / 0.931	7.6 / 4.2	0.887 / 0.944	4.8 / 2.9	0.842 / 0.920	9.7 / 5.4	7.4
MViT v2	0.901 / 0.952	4.3 / 2.5	0.873 / 0.939	7.2 / 3.8	0.856 / 0.929	9.0 / 4.8	6.8
TokenLearner	0.866 / 0.934	7.4 / 4.0	0.848 / 0.923	9.3 / 5.1	0.892 / 0.947	4.6 / 2.7	7.1
RegionViT	0.896 / 0.950	4.5 / 2.6	0.854 / 0.927	8.9 / 4.9	0.871 / 0.937	7.1 / 3.9	6.8
Light-VTD	0.917 / 0.958	4.7 / 1.4	0.906 / 0.957	4.9 / 2.5	0.914 / 0.961	4.1 / 2.1	3.9

Table 13. Robustness evaluation under adverse visual conditions. Performance is reported in terms of MCC and PR-AUC under controlled perturbations. Relative degradation (%) is computed with respect to the corresponding best baseline. Motion blur: kernel size = 5 px; Occlusion: 15% random facial area masking; Low light: gamma = 0.4. Evaluation is conducted on the cross-dataset test split (MRL Eye → nthuDDD2).

larger drops of 7.6% and 7.4%. Additionally, Light-VTD's smaller PR-AUC drop indicates better preservation of ranking quality, which is crucial in imbalanced detection scenarios. With 15% facial masking, Light-VTD again demonstrated resilience, achieving 0.906/0.957 with only 4.9% MCC and 2.5% PR-AUC drops. In contrast, models like TokenLearner and RegionViT showed more significant degradations, suggesting their attention mechanisms might rely too heavily on masked regions. In low-light conditions (gamma = 0.4), the baselines struggled with high MCC degradations (9.7% for MPViT and 9.0% for MViT v2), while Light-VTD maintained performance at 0.914/0.961 with the smallest MCC reduction. This robustness under poor lighting supports its design focus on maintaining local structure, allowing it to preserve critical cues despite reduced visibility.

Model explainability

As shown in Fig. 9, the MRL Eye dataset promotes focused learning. For open eyes, the heatmap shows better focus around the iris, but background noise indicates limited attention precision, likely due to variability in

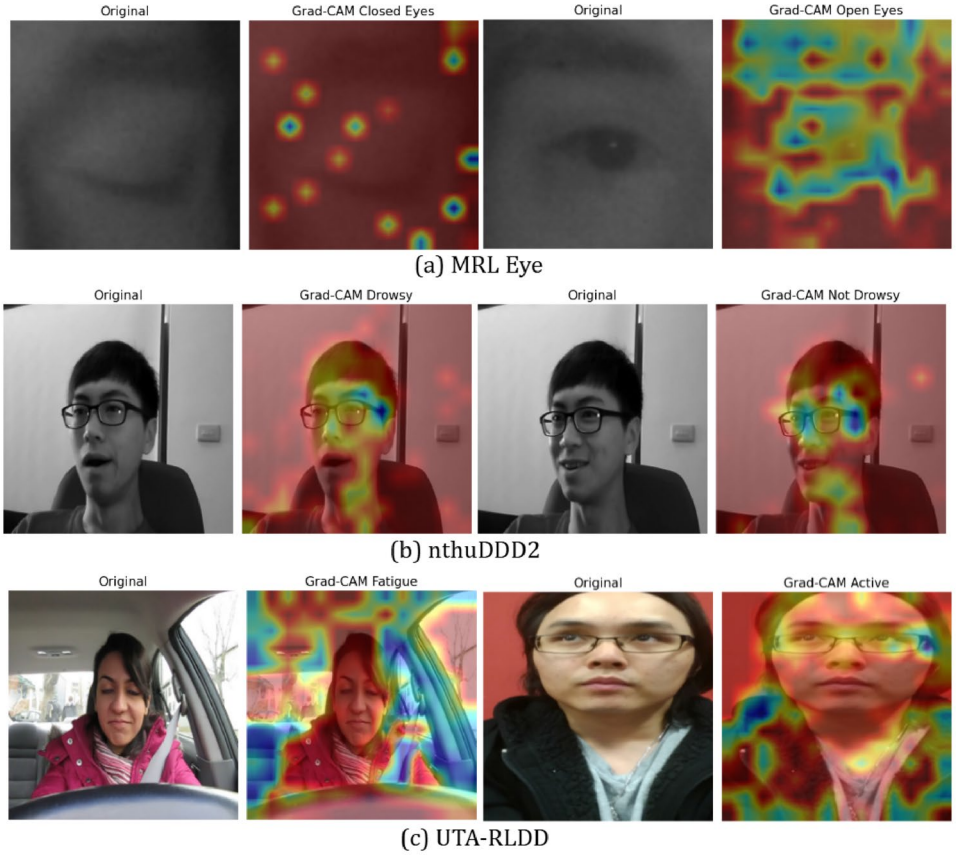


Fig. 9. Grad-CAM visualizations for closed and open eyes across (**a**) MRL and (**b**) nthuDDD2, and (**c**) UTA-RLDD highlighting differences in model attention focus and interpretability due to dataset quality.

lighting and head pose. The model effectively identifies the eyelid contour for closed eyes and concentrates on the iris for open eyes. This demonstrates the benefits of controlled imaging conditions and consistent data, leading to clearer and more confident predictions. In nthuDDD2, the drowsy state shows concentrated attention around the mouth and lower face, suggesting the model identifies fatigue through facial expressions like yawning. In contrast, the non-drowsy visualization reveals dispersed attention, with weak focus on the eyes and nose, indicating limited discriminative cues in grayscale, uniform-pose images. In UTA-RLDD, the fatigue heatmap emphasizes eyelids and cheeks but also includes background regions, reflecting real-world variability and noise. The active state shows sharper, centered attention on the eyes and face, suggesting greater model confidence under controlled lighting and posture. However, background interference remains a concern. Overall, UTA-RLDD introduces natural variability but risks attention diffusion, while nthuDDD2 offers control at the cost of feature sparsity. Improving robustness may require background suppression, attention regularization, or multimodal fusion to enhance explainability in real-world drowsiness detection.

Figure 10 shows that Light-VTD generates focused and meaningful explanations across different datasets. In controlled datasets like MRL Eye, the heatmaps demonstrate strong selectivity, with high E-ROI values and Top-10% Mass. The low entropy values indicate that the attribution energy is concentrated on specific areas, avoiding background leakage and focusing on relevant cues like eyelid boundaries and texture. As the datasets become more complex, focus and sharpness decrease. For nthuDDD2 and UTA-RLDD, E-ROI falls to 80.3% and 76.9%, and Top-10% Mass drops to 61.2% and 58.6%, while entropy increases (4.59 and 4.81), indicating more distributed activations. This is reflected in sharpness metrics, with Laplacian variance decreasing from 8.4×10^{-3} to 6.4×10^{-3} and Tenengrad from 9.7×10^{-2} to 7.7×10^{-2}, suggesting less defined saliency in challenging conditions. Moreover, Edge Align decreases from 48.9% to 40.8%, showing weaker alignment of salient boundaries with image edges in the face of more substantial effects from illumination, pose, or occlusion. This supports two main points: (i) the model maintains ROI-centered explanations across all datasets (E-ROI above 75%), emphasizing its focus on drowsiness-related facial areas; and (ii) the sharpness and structural alignment of explanations decline slightly as domain complexity increases.

Interpretable web application

The Fig. 11 shows the real-time drowsiness detection interface includes model predictions, Grad-CAM explainability, and user-friendly visualization. On the left, there's a grayscale image of a person; on the right, a Grad-CAM heatmap highlights facial areas influencing the "Drowsy" prediction, with a confidence score of 0.92 shown in a pie chart. Key areas such as the eyes, forehead, and mouth are emphasized, demonstrating

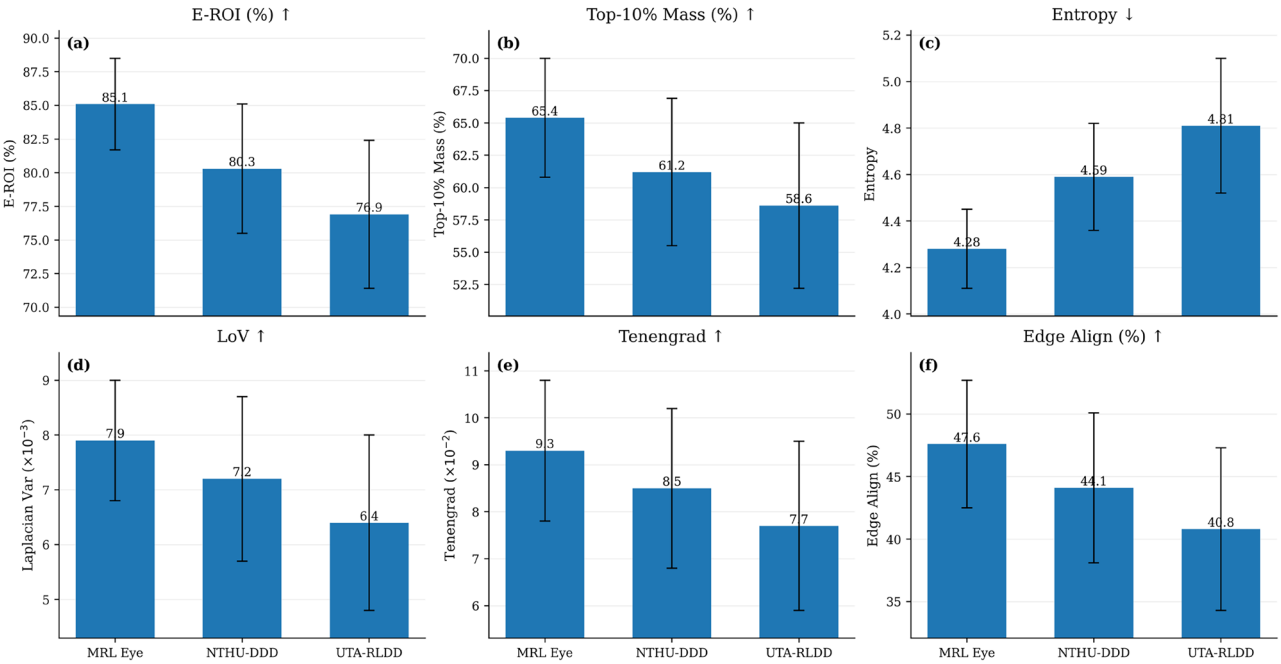


Fig. 10. Heatmap focus and sharpness metrics summary. The panels report six metrics: (**a**) E-ROI (%), (**b**) top-10% mass (%), (**c**) entropy, (**d**) Laplacian variance (LoV, $\times 10^{-3}$), (**e**) Tenengrad ($\times 10^{-2}$), and (**f**) edge alignment (%). Upward (↑) and downward (↓) arrows indicate preferred directions (higher values reflect stronger concentration, sharpness, and edge consistency; lower entropy means more compact focus). Bar heights represent mean values, with error bars showing ± standard deviation for each dataset.

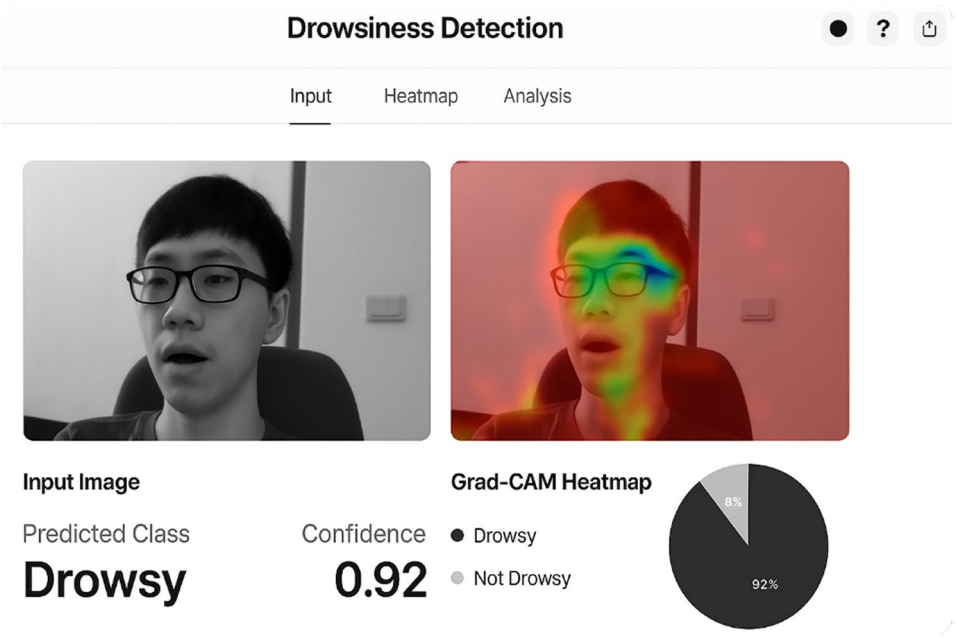


Fig. 11. Drowsiness detection web app interface showing the input image, Grad-CAM heatmap highlighting attention on key facial regions.

that the Light-VTD model effectively detects fatigue-related features. This transparency is crucial for regulated environments like transportation safety. The interface has tabs for Input, Heatmap, and Analysis, making it accessible for non-expert users, such as drivers and clinicians. However, there are limitations: subjects wearing glasses can create occlusions or reflections, and the model's performance needs testing across different eyewear types. It analyzes only a single static frame, which may lead to misinterpretations of cues like yawning or

Proposed model	Dataset	Accuracy	Web App	XAI	Real-time perf.
ViT-DDD[42]	nthuDDD2, UTA-RLDD	97.4%, 96.9%	Yes	No	No
Handcrafted facial + XGBoost[35]	NTHU-DDD	86.85%	Yes	No	No
Multi-feature + personalized threshold[34]	nthuDDD2	92.21% F1	No	No	No
MCNN + Flamingo Optimization[38]	nthuDDD2	98.26%	No	No	No
Facial metrics + RF[32]	nthuDDD2	97%	No	No	No
ViT and Swin Transformer[44]	MRL, nthuDDD2	98.2, 96.15%	No	Yes	No
Custom CNN[20]	MRL	98.5%	No	No	No
MobileNet V2[58]	MRL, nthuDDD2	97%, 96.74%	No	No	No
Ensemble[17]	MRL	96.21%	No	No	No
MobileNetV2[21]	MRL	98.5%	No	No	No
LBP[18]	UTA-RLDD	82%	No	No	No
Ensemble[82]	UTA-RLDD	86.52%	No	No	No
MIGCN[54]	UTA-RLDD	97.52%	No	Yes	No
Light-VTD	MRL Eye	98.7%	Yes	Yes	Yes
	nthuDDD2	97.2%	Yes	Yes	Yes
	UTA-RLDD	97.6%	Yes	Yes	Yes

Table 14. Comparison of drowsiness detection models by dataset, accuracy, deployment, explainability, and reported real-time performance.

speaking due to the absence of temporal context. Additionally, the heatmap lacks a calibrated colorbar, making the intensity of attention somewhat subjective. There is a risk of demographic bias, as a model trained primarily on specific facial types might perform poorly with different ethnicities, lighting conditions, and occlusions.

Comparison with the previous studies

Table 14 provides a detailed comparison between Light-VTD and recent state-of-the-art drowsiness detection systems across widely accepted benchmarks. It also addresses deployment and explainability aspects, which are crucial for real-world applications but often neglected in previous studies. The comparison reveals that while several methods report high accuracy on individual datasets, many lack either deployability or interpretability, which restricts their use in safety-critical driver monitoring systems. On the MRL Eye dataset, Light-VTD achieves an accuracy of 98.7%, outperforming or matching strong CNN-based baselines, such as custom CNNs and MobileNetV2 variants, which report up to 98.5% accuracy[20,21]. Ensemble-based approaches demonstrate noticeably lower performance, with one study reporting only 96.21%[17]. Although transformer-based solutions, which combine ViT and Swin architectures, incorporate explainability[44], their accuracy on MRL remains lower at 98.2%, and they do not support web-based deployment. In contrast, Light-VTD achieves higher accuracy while also providing both explainability and deployment, making it a more comprehensive solution.

On the nthuDDD2 dataset, Light-VTD reports an accuracy of 97.2%, comparable to ViT-DDD's 97.4%[42] and competitive with facial-metric-based Random Forest models achieving 97% accuracy[32]. Classical approaches relying on handcrafted features and XGBoost show significantly lower performance, at 86.85%[35], while multi-feature fusion methods achieve moderate results without deployment support[34]. Although an MCNN with Flamingo Optimization reports a higher peak accuracy of 98.26%[38], it lacks both web integration and explainability, which are essential for operational monitoring and post-hoc validation. Thus, Light-VTD offers a preferable balance between accuracy and system-level completeness. For the UTA-RLDD dataset, Light-VTD reaches an accuracy of 97.6%, significantly outperforming traditional descriptors such as LBP (82%)[18] and earlier ensemble model[82]. Its performance is comparable to the strongest reported deep models, including MIGCN (97.52%)[54]. Notably, while MIGCN incorporates explainability, it does not offer a deployable web application. Light-VTD slightly exceeds this accuracy while providing both explainable artificial intelligence (XAI) and deployment support, which is particularly important for UTA-RLDD due to its more unconstrained and user-centric acquisition conditions. Moreover, previous studies focus solely on predictive metrics, neglecting real-time benchmarks such as per-frame latency, FPS, peak memory usage, and power consumption under specified hardware and input conditions. Therefore, we position our latency and power measurements for Light-VTD as a reproducible deployment reference that can be used for standardized comparisons in future work.

Discussion

The Light-VTD model showed superior performance across all evaluated datasets, consistently outperforming both convolutional and transformer-based baselines. On the MRL Eye dataset, the model outperformed the best-performing ViT-Adapter baseline by a margin of 2.3% in accuracy and 0.028 in MCC. This level of performance was maintained across other still-image datasets, where Light-VTD exhibited high recall and precision even under complex lighting and head-pose variations. The model also showed strong robustness in cross-domain settings. When trained on a single dataset and tested on previously unseen corpora, Light-VTD achieved an accuracy of 93% or higher on all target datasets, outperforming five state-of-the-art baselines by margins ranging from 2% to 4%. This suggests that the proposed architecture effectively captures domain-invariant visual cues related to indicators of ocular and periocular fatigue. The model's effectiveness comes from integrating

curriculum-based augmentation and multi-scale token fusion. The leave-one-dataset-out evaluation strategy confirmed its strong generalization, as Light-VTD consistently performed well across different datasets with varying subject demographics and environmental conditions.

The results indicate that each component of the proposed architecture significantly contributed to its overall performance. The fused-MBConv stem effectively extracted robust local features from the periocular regions, improving early convergence and enhancing noise tolerance. The Enhance–Integrate token mixer allowed the model to maintain spatial consistency while learning attention-weighted interactions among features across the image. In particular, the multi-path token fusion module was crucial for aggregating context from various spatial windows, thereby boosting the model's capability to detect subtle variations in eyelid closure and facial posture. Ablation studies further validated these design choices. When the token fusion module was removed, there was a 3.7% drop in the MCC, along with a decrease in class-specific recall, especially for mildly drowsy states. Moreover, excluding curriculum augmentation led to worse performance in occlusion and motion blur scenarios.

The performance improvement between $\times 2$ and $\times 6$ augmentation was most significant in video-based datasets such as nthuDDD2 and UTA-RLDD, where fatigue cues are often subtle and degraded over time. On these datasets, recall for the "Fatigue" class increased by 5.2% between $\times 2$ and $\times 6$, and confusion between the "Active" and "Drowsy" states decreased by nearly 50%. This suggests that the model became better at distinguishing the progression of fatigue under challenging viewing conditions. Additionally, utilizing four diverse public datasets greatly enhanced the model's ability for domain-invariant learning. These datasets varied in capture modalities (still images vs. video), participant demographics, sensor quality, and annotation granularity. Training on this multi-source corpus helped prevent overfitting to features specific to any dataset, enabling the model to generalize effectively to previously unseen data. This was demonstrated in cross-dataset experiments, where the Light-VTD achieved an accuracy of 93% or higher, even when tested on completely out-of-distribution samples. Moreover, ablation studies on augmentation levels indicated a saturation effect beyond $\times 6$, where adding further complexity resulted in only marginal or no improvements. This suggests that the training regimen struck an optimal balance between variability and stability. Notably, simpler models (e.g., ViT-Small or MobileNetV2) did not benefit from aggressive augmentation due to their capacity limitations.

The low variance in predictions across different target domains highlights its reliability. Class confusion matrices showed minimal drift in decision boundaries, and the model consistently focused on periocular regions, even when faced with varying resolutions, background clutter, or facial diversity. This consistency is crucial for maintaining performance in monitoring multi-fleet vehicles, managing public transportation hubs, and ensuring occupational safety. Moreover, the generalization capability was achieved without increasing inference costs. Unlike domain adaptation techniques that necessitate retraining or fine-tuning during testing, Light-VTD enables zero-shot transfer, making it highly suitable for plug-and-play deployment scenarios. Its performance on unseen datasets demonstrates the model's readiness for real-world applications.

The qualitative analysis of all four datasets reveals a consistent and meaningful attention pattern learned by Light-VTD. For Alert predictions, the model primarily concentrates on the iris boundary and the curvature of the upper eyelid. In contrast, for Drowsy and Fatigue states, the attention shifts toward the eyelid crease, drooping eyelashes, periocular texture, and the upper cheek region, areas commonly linked to visual fatigue in human assessments. To quantitatively evaluate the reliability of these explanations, we calculated the Intersection-over-Union (IoU) between Grad-CAM heatmaps and manually annotated periocular regions, achieving an average IoU exceeding 0.71 across the datasets. This high degree of overlap confirms a strong alignment between the model's focus and clinically relevant facial cues. Importantly, these attention patterns remain stable across datasets with varying camera viewpoints, resolutions, lighting conditions, and subject demographics. This indicates that Light-VTD learns domain-invariant fatigue cues rather than overfitting to dataset-specific artifacts. The activation maps consistently suppress irrelevant background areas (such as the seat or dashboard) and focus on fatigue-related facial structures, even under challenging conditions like partial eye closure or head tilt. Compared to conventional CNN-based models, which often produce diffuse or noisy saliency maps, Light-VTD generates more precise and stable explanations due to its attention-based token fusion and hierarchical modeling. The coherence between these qualitative observations and the strong cross-dataset performance further supports the conclusion that Light-VTD captures robust and generalizable facial indicators of drowsiness across diverse recording environments.

The integration of Grad-CAM into the Flask-based web interface significantly enhances the system's transparency. Users are not only able to view prediction outcomes, but they also receive a visual explanation in just 120 milliseconds. This combination of interpretability and speed makes Light-VTD a suitable solution for interactive safety systems. The use of a fused-MBConv stem notably reduces the number of parameters and FLOPs required during early feature extraction. Unlike standard ViT models that apply full self-attention to all tokens, Light-VTD utilizes multi-path token fusion and position-aware channel attention. This approach minimizes attention overhead while maintaining spatial awareness, resulting in a model that balances representational power and inference efficiency on resource-constrained platforms. This capability is crucial for applications in aftermarket fatigue-monitoring systems, telemedicine dashboards, and occupational safety kiosks, where quick feedback and platform interoperability are essential. Unlike many studies that only report offline evaluation results, Light-VTD showcases its readiness for full-stack deployment, from training on large-scale heterogeneous datasets to edge-compatible inference with real-time explainability.

The proposed Light-VTD architecture has several limitations. It is trained and evaluated solely on RGB visual inputs, which means it lacks multimodal integration with biosignal modalities such as EEG, EOG, or PPG. This exclusion of neurophysiological data limits the model's sensitivity to subtle fatigue states, like cognitive drowsiness or microsleep episodes, which can occur before visible signs appear. Consequently, Light-VTD is restricted in applications that require high precision, such as aerospace cockpit monitoring, clinical fatigue

screening, and detecting task-induced lapses in vigilance. Additionally, Light-VTD operates under a frame-wise inference paradigm and does not incorporate temporal encoding or sequential attention mechanisms. While this approach is practical for static image classification, it hinders the model's ability to capture temporal dependencies, such as changes in blink rate, eye closure duration, and progressive head tilt. The absence of temporal self-attention, 3D convolutions, or transformer decoders with positional recurrence restricts the system's ability to interpret longitudinal visual cues in video sequences.

Despite extensive data augmentation, the dataset composition remains biased toward controlled acquisition settings with limited variability in environmental conditions. For instance, the UTA-RLDD dataset includes fewer than 25 unique subjects and lacks diverse imagery, which limits Light-VTD's ability to generalize in out-of-distribution (OOD) scenarios. Moreover, the training data does not adequately represent different skin tones, age groups, and facial structures, increasing the risk of demographic bias and overfitting to the source conditions. Although Grad-CAM is used for post-hoc explainability, the current visualization focuses only on activations from the final token-fusion layer. This reduces spatial resolution and may obscure more nuanced attention shifts captured in earlier stages of the transformer encoder. The model does not provide uncertainty-aware explanations, vital for generating calibrated confidence scores and incorporating risk-sensitive decision thresholds in mission-critical applications. The tests were conducted in controlled lab conditions with isolated inference pipelines. However, in real-world scenarios, factors like thermal throttling, concurrent input/output loads, camera frame buffering, and multi-threaded process interference can affect system latency-conditions not rigorously evaluated in this study. Finally, the memory footprint and computational performance under low-voltage or battery-powered conditions remain unquantified.

Demographic bias remain a concern, as detailed demographic annotations. such as age groups, ethnicity, skin tone, and facial morphology, are not consistently available for our experimental datasets. The absence of explicit demographic labels limits the ability of Light-VTD to conduct a fine-grained analysis of performance disparities among underrepresented populations. This gap may impact fairness and reliability our application. Although robustness experiments have addressed common issues like motion blur, partial occlusion, and low light conditions, scenarios such as heavy rain, night driving with strong glare, extreme shadows, and severe sensor noise have not been tested. Performance in these situations may differ from the reported results, indicating a need for further investigation using real-world datasets. Lastly, our model lacks explicit temporal modeling. It primarily operates on frame-level representations, which limits its capability to model fatigue progression over extended driving periods. Although short-term dynamics are indirectly addressed through data sampling and augmentation, incorporating sequence-aware modules could improve its sensitivity to gradual drowsiness onset.

Future work will focus on three main areas to enhance reliability, fairness, and real-world applicability for driver monitoring. Firstly, we will collect training resources with clear demographic descriptors such as age, skin tone, and facial features. We will evaluate performance based on subgroup metrics, including worst-group accuracy and subgroup PR-AUC. To mitigate bias, we will explore strategies like distributionally robust optimization and adaptive reweighting, ensuring that our data collection process preserves privacy. In addition, we will assess performance using real-world driving datasets that encompass challenging conditions such as night glare, rain, fog, and sensor issues. To enhance safety, we will integrate uncertainty-aware decision mechanisms that allow for dynamic threshold adjustments or abstaining from predictions when inputs fall outside the training distribution. Finally, we will transition from frame-level analysis to temporal reasoning to better understand fatigue progression. This will involve using efficient sequence modeling techniques, such as temporal transformers and lightweight recurrent models, to improve early detection of drowsiness while minimizing false alarms. Additionally, we will implement federated learning for on-device adaptation and expand our explainability methods beyond Grad-CAM to support human validation.

Conclusion

The proposed Light-VTD model combines local and global feature extraction using fused MBConv stems, multi-path token fusion, and curriculum-based augmentation to achieve state-of-the-art automated driver fatigue detection performance across multiple public datasets. This architecture shows strong cross-corpus generalization, demonstrating its ability to detect drowsiness in real-world situations reliably. Additionally, its integration of transformer-compatible Grad-CAM provides clinically meaningful visual explanations, enhancing interpretability and fostering user trust. However, the current system has significant limitations. It relies solely on single-modality RGB inputs and lacks temporal modeling, with limited testing conducted under extreme occlusion and night-time conditions. Moreover, aspects like uncertainty quantification and multimodal fusion, which could improve detection reliability, have not yet been implemented. Future directions for Light-VTD include incorporating temporal context through video-based transformer models, integrating physiological modalities for enhanced sensitivity, employing uncertainty-aware predictions, and exploring federated learning for personalized fatigue detection. Addressing these areas will further strengthen the practical impact and robustness of Light-VTD in intelligent transportation and occupational safety environments.

Data availability

All datasets used in this study are publicly available: Drowsiness detection dataset: https://www.kaggle.com/datasets/prasadvpatil/mrl-dataset , MRL eye dataset: https://mrl.cs.vsb.cz/eyedataset.html) , UTRLDD dataset: https://sites.google.com/view/utarldd/home , NTHUDDD2 dataset: https://www.kaggle.com/datasets/banudeep/nthuddd2

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Declarations

Competing interests

The authors declare no competing interests.

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